

**INTEGRATIVE TRANSCRIPTOMIC AND MULTI-OMICS ANALYSIS
OF SHARED INFLAMMATORY MECHANISMS IN ALLERGIC
RESPIRATORY DISEASES**

Submitted in partial fulfilment of the required for the award of
Master of Science Degree in Bioinformatics and Data Science

By

CHARULATHA.M (44739004)



**DEPARTMENT OF BIOINFORMATICS
SCHOOL OF SCIENCE AND HUMANITIES**

SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
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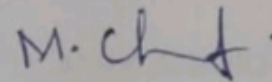
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DECLARATION

I, **CHARULATHA M (44739004)** hereby declare that the project report titled **INTEGRATIVE TRANSCRIPTOMIC AND MULTI-OMICS ANALYSIS OF SHARED INFLAMMATORY MECHANISMS IN ALLERGIC RESPIRATORY DISEASES** done by me under the guidance of **Dr. D. ALEX ANAND** at Sathyabama Institute of Science and Technology, Jeppiar Nagar, Rajiv Gandhi Road, Chennai-600119, Tamil Nadu, India is submitted in partial fulfilment of the requirements for the award of Master of Science degree in Bioinformatics and Data Science.

DATE: 28. 04. 2026

PLACE: CHENNAI



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CHARULATHA. M

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **Ms. Charulatha. M (44739004)** who carried out the final year project titled **INTEGRATIVE TRANSCRIPTOMIC AND MULTI-OMICS ANALYSIS OF SHARED INFLAMMATORY MECHANISMS IN ALLERGIC RESPIRATORY DISEASES** under the supervision of **DR. D. ALEX ANAND** from **DECEMBER 2026** to **APRIL 2026**.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Allergic respiratory diseases — dust allergy, chronic rhinosinusitis with nasal polyps (CRSwNP), and allergic asthma — are chronic inflammatory conditions that share common immunological features yet have rarely been investigated together at the molecular level. This study aimed to systematically identify shared molecular pathways across all three conditions through pan-transcriptomic analysis, and to validate those findings at the protein and metabolite levels using sinusitis multi-omics data.

Three publicly available GEO microarray datasets — GSE9150, GSE23552, and GSE65204 — were processed using a standardised R pipeline. Cross-disease gene overlap identified 15 shared discovery genes and 4 strict high-confidence genes. Functional enrichment analysis using Metascape, g:Profiler, and Enrichr identified 9 consensus genes. Protein–protein interaction network analysis in STRING and Cytoscape using four CytoHubba centrality measures identified four hub genes: CCL5, CCL20, IL7R, and IDO1. Sinusitis multi-omics data from published proteomic and metabolomic studies were integrated using a Python-based cross-omics pathway scoring framework. Two-step biomarker validation was conducted using recent literature and three independent external datasets. Machine learning evaluation using Random Forest and Logistic Regression with five-fold cross-validation confirmed the discriminative potential of the candidate panel.

Results showed that immune activation, cytokine signalling, neutrophil-mediated innate immunity, and chemokine-driven leukocyte recruitment are the dominant shared biological themes across all three diseases, supported consistently across transcriptomic, proteomic, and metabolomic levels. CCL20 and IDO1 were validated in two of three external datasets. The machine learning Part 2 biomarker panel achieved AUC values of 0.776 (Random Forest) and 0.816 (Logistic Regression). These findings demonstrate that integrative multi-omics analysis provides a more biologically meaningful approach to biomarker identification in allergic respiratory disease than single-omics methods alone.

Keywords: allergic respiratory diseases; transcriptomics; multi-omics integration; differential gene expression; functional enrichment; protein–protein interaction; hub genes; biomarker discovery; CCL5; CCL20; IL7R; IDO1; machine learning; cytokine signalling.

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LIST OF ABBREVIATIONS

AhR	Aryl Hydrocarbon Receptor
ALDH1A3	Aldehyde Dehydrogenase 1 Family Member A3
AUC	Area Under the Curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BH	Benjamini–Hochberg (multiple testing correction)
CASP4	Caspase 4
CCL5	C-C Motif Chemokine Ligand 5 (RANTES)
CCL20	C-C Motif Chemokine Ligand 20 (MIP-3 α)
CCR1	C-C Chemokine Receptor Type 1
CCR3	C-C Chemokine Receptor Type 3
CCR5	C-C Chemokine Receptor Type 5
CCR6	C-C Chemokine Receptor Type 6
CD36	Cluster of Differentiation 36 (Scavenger Receptor)
CD127	Cluster of Differentiation 127 (IL-7 Receptor Alpha Chain)
CORO1A	Coronin Actin Binding Protein 1A
CRS	Chronic Rhinosinusitis
CRSwNP	Chronic Rhinosinusitis with Nasal Polyps
CTSC	Cathepsin C
DEG	Differentially Expressed Gene
DGE	Differential Gene Expression
DisGeNET	Disease-Gene Network Database
EAACI	European Academy of Allergy and Clinical Immunology

eBayes	Empirical Bayes (variance moderation in limma)
EGFL6	EGF-Like Domain Multiple 6
ELISA	Enzyme-Linked Immunosorbent Assay
ELOVL5	Elongation of Very Long Chain Fatty Acids Protein 5
Enrichr	Enrichment Analysis Web Server (Ma'ayan Laboratory)
F1	F1 Score (harmonic mean of precision and recall)
FC	Fold Change
GBD	Global Burden of Disease
GCN2	General Control Non-derepressible 2 Kinase
GEO	Gene Expression Omnibus (NCBI repository)
GO	Gene Ontology
GPL	Gene Platform (GEO platform identifier)
GSE9150	GEO Series 9150 — Dust Allergy Dataset
GSE23552	GEO Series 23552 — CRSwNP Dataset
GSE65204	GEO Series 65204 — Allergic Asthma Dataset
GSE72713	GEO Series 72713 — External Validation Dataset
GSE75011	GEO Series 75011 — External Validation Dataset
GSE179265	GEO Series 179265 — External Validation Dataset
g:Profiler	Gene Profiler (Functional Enrichment Tool)
g:SCS	Set Counts and Sizes (g:Profiler multiple testing correction)
HDM	House Dust Mite
HGNC	HUGO Gene Nomenclature Committee
HMDB	Human Metabolome Database

IDO1	Indoleamine 2,3-Dioxygenase 1
IFN- γ	Interferon Gamma
IgE	Immunoglobulin E
IL-4	Interleukin 4
IL-4R α	Interleukin 4 Receptor Alpha Subunit
IL-5	Interleukin 5
IL-6	Interleukin 6
IL-7	Interleukin 7
IL-8	Interleukin 8
IL-13	Interleukin 13
IL-25	Interleukin 25
IL-33	Interleukin 33
IL7R	Interleukin 7 Receptor (Gene)
ILC2	Group 2 Innate Lymphoid Cells
INTS10	Integrator Complex Subunit 10
JAK-STAT6	Janus Kinase — Signal Transducer and Activator of Transcription 6
KEGG	Kyoto Encyclopedia of Genes and Genomes
LAMB3	Laminin Subunit Beta 3
LC-MS/MS	Liquid Chromatography Tandem Mass Spectrometry
log2FC	Log2 Fold Change
LR	Logistic Regression
MAOA	Monoamine Oxidase A

MCC	Maximal Clique Centrality
MetaboAnalyst	Metabolomics Data Analysis Web Server
Metascape	Multi-Omics Pathway Enrichment Platform
ML	Machine Learning
mRNA	Messenger Ribonucleic Acid
NCBI	National Center for Biotechnology Information
NK	Natural Killer (Cells)
OMIM	Online Mendelian Inheritance in Man
PAG1	Phosphoprotein Membrane Anchor with Glycosphingolipid Microdomains 1
PCA	Principal Component Analysis
PPI	Protein–Protein Interaction
RANTES	Regulated on Activation, Normal T-cell Expressed and Secreted (CCL5)
Reactome	Reactome Pathway Knowledgebase
RF	Random Forest
RMA	Robust Multi-array Average
RNA-seq	RNA Sequencing
ROC	Receiver Operating Characteristic
STRING	Search Tool for the Retrieval of Interacting Genes/Proteins
Th2	T Helper 2 Cells
TNF	Tumor Necrosis Factor
TSLP	Thymic Stromal Lymphopoietin
WikiPathways	Community-Curated Biological Pathways Database

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CHAPTER 1

INTRODUCTION

Allergic respiratory diseases including dust allergy, chronic rhinosinusitis with nasal polyps (CRSwNP), and allergic asthma are amongst the leading causes of chronic non-communicable disease globally, collectively affecting millions across all age groups. These conditions are driven by dysregulated type 2 immune responses, IgE-mediated sensitisation, eosinophilic inflammation, and cytokine-mediated mucosal remodelling, and frequently co-occur in the same patients — a pattern consistent with the united airways concept (Bousquet *et al.*, 2019; Shin *et al.*, 2023). Despite this clear biological overlap, most existing studies examine each disease in isolation and at a single molecular level, leaving the shared molecular mechanisms across these conditions poorly defined.

This study presents an integrative analysis of common molecular mechanisms across dust allergy, sinusitis, and allergic asthma. Pan-transcriptomic analysis of publicly available microarray datasets (GSE9150, GSE23552, and GSE65204) from the NCBI GEO database was performed using a standardised R pipeline, followed by cross-disease gene overlap, functional enrichment analysis, and protein–protein interaction (PPI) network analysis to identify shared hub genes. For multi-omics validation, sinusitis was selected as a representative disease based on the availability of published proteomic and metabolomic datasets, and a Python-based quantitative cross-omics pathway integration framework was applied. Machine learning classifiers were developed to evaluate the discriminative performance of the identified biomarkers.

1.1 BACKGROUND AND CLINICAL SIGNIFICANCE

Dust allergy, CRSwNP, and allergic asthma collectively represent one of the most prevalent categories of chronic non-communicable disease worldwide, with rising prevalence driven by urbanisation, air pollution, and changing allergen exposure patterns (Shin *et al.*, 2023; GBD 2021 Allergic Disease Collaborators, 2025). All three conditions are driven by dysregulated type 2 immune responses and frequently co-occur in the same patients — a pattern consistent with the united airways concept, which proposes that the upper and lower respiratory tracts form a

single immunological unit (Bousquet *et al.*, 2019). Despite this shared pathophysiological foundation, molecular research continues to treat these diseases as separate entities, with few studies systematically comparing shared transcriptional and multi-omics signatures across all three conditions simultaneously.

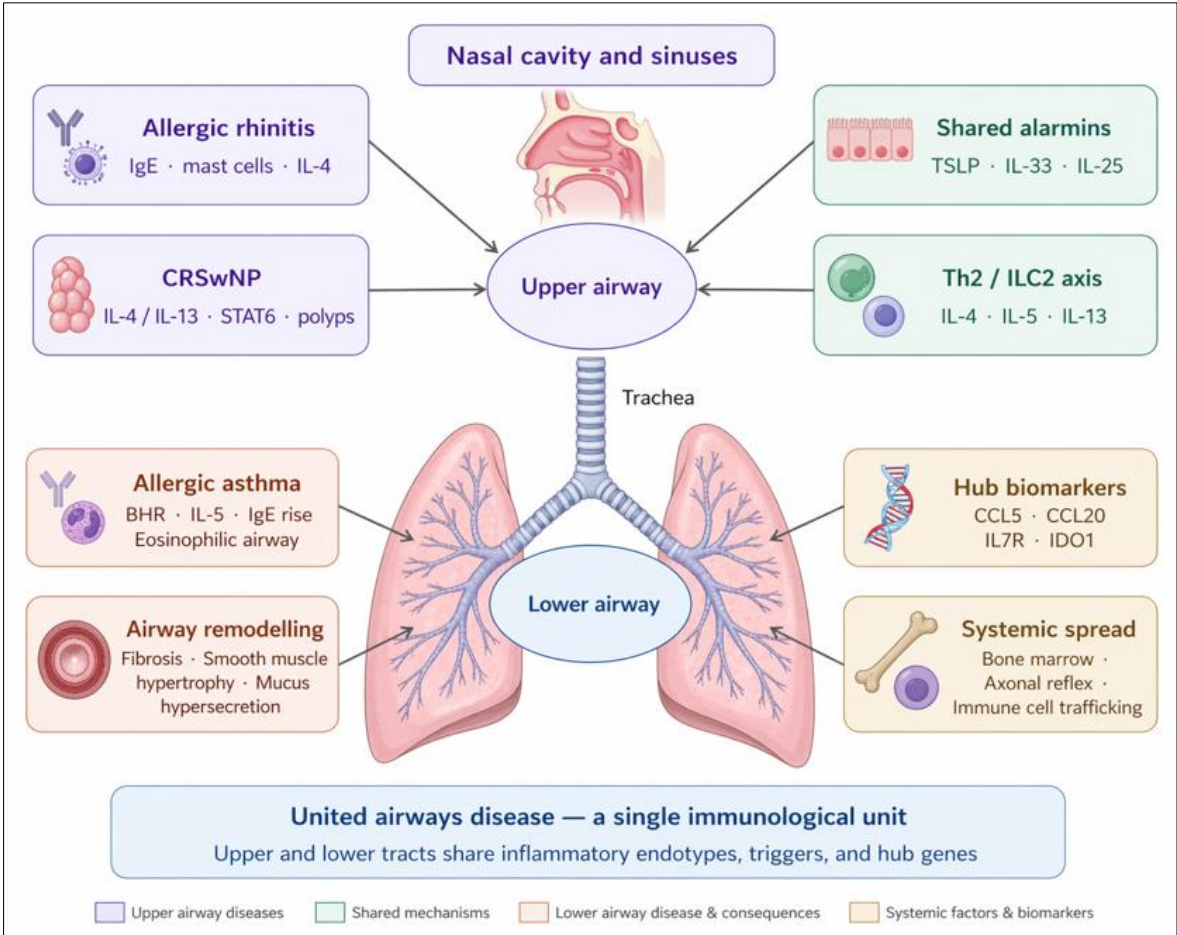


Fig:1.1: United Airways Disease concept

The united airway concept: upper and lower respiratory tracts as a single immunological unit (Bousquet *et al.*, 2019; Shin *et al.*, 2023; Iyer *et al.*, 2024).

1.2 GAP IN KNOWLEDGE AND RESEARCH RATIONALE

Three interconnected gaps motivate this research. First, systematic cross-disease transcriptomic comparison of dust allergy, CRSwNP, and allergic asthma using reproducible analytical frameworks is rare. Most studies examine a single disease, a single dataset, or a single tissue type, limiting the identification of shared disease-driving mechanisms (Rosati *et al.*, 2024). Second, transcriptomic findings in allergic respiratory diseases have rarely been validated at the protein and metabolite levels, preventing confident assignment of functional biological relevance to candidate biomarkers (Chen *et al.*, 2023; Subramanian *et al.*, 2020). Third, the absence of integrative multi-omics analysis limits the ability to identify mechanistically credible biomarker candidates relevant across the full allergic disease continuum. This study directly addresses all three gaps by combining pan-transcriptomic discovery, multi-omics validation in sinusitis, and machine learning-based biomarker evaluation within a single reproducible computational pipeline (Agache *et al.*, 2025; Jutel *et al.*, 2023).

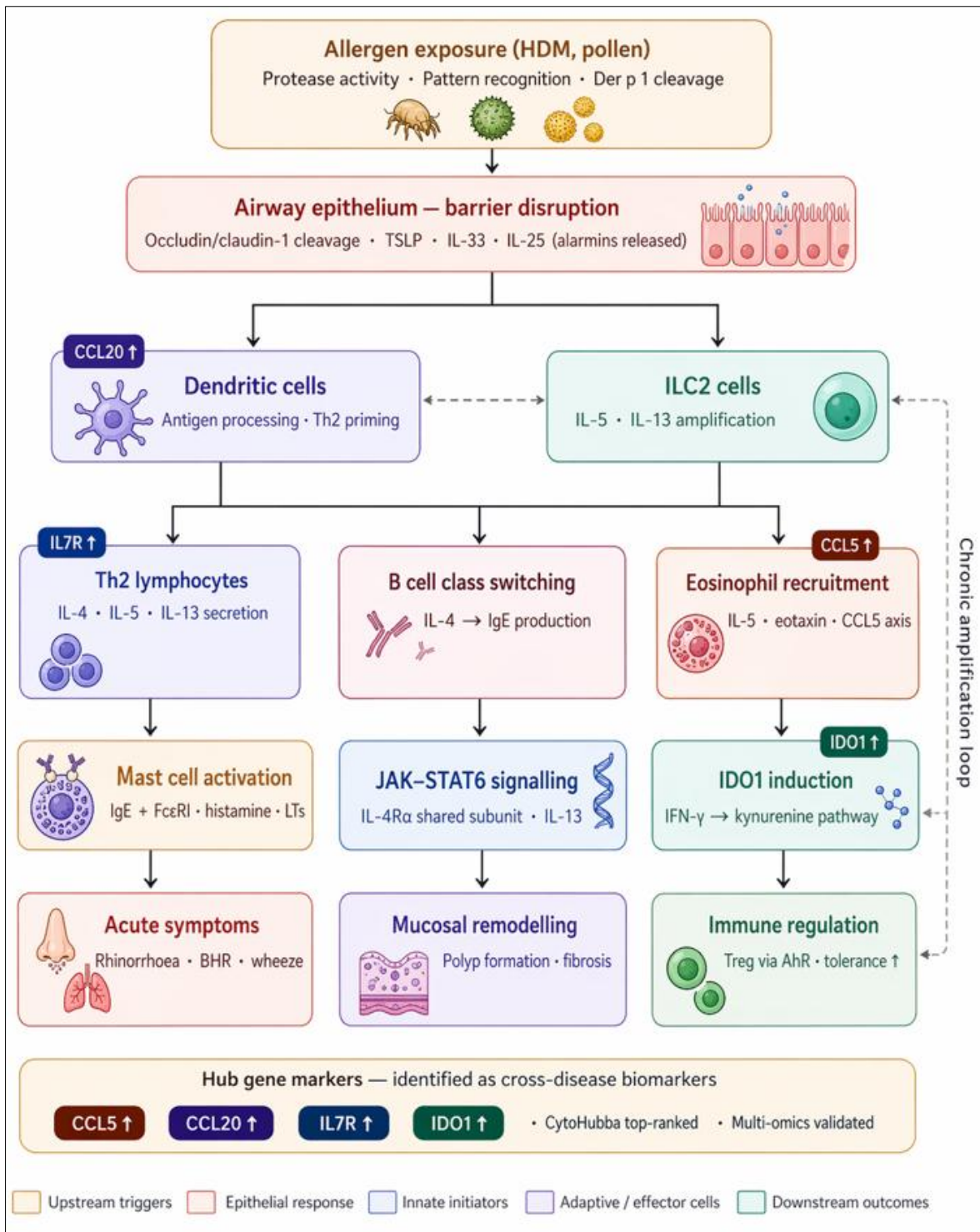


Fig: 1.2: Type 2 Inflammatory Pathway

Type 2 inflammatory and immune response pathway in allergic respiratory disease with hub gene annotations (Ha & Kim, 2025; Iyer *et al.*, 2024; Dorismond *et al.*, 2025; Le Mai *et al.*, 2020; Miao *et al.*, 2024).

1.3 TRANSCRIPTOMICS AS A DISCOVERY PLATFORM

Transcriptomics, the genome-wide measurement of RNA abundance, allows simultaneous examination of thousands of genes from diseased and healthy samples. Microarray data stored in public repositories such as NCBI GEO — with over 225,000 deposited datasets — enable large-scale integrative studies without additional experimental costs (Clough & Barrett 2016). Differential gene expression (DGE) analysis using a standard pipeline of background correction, RMA normalisation, PCA-based quality control, and limma-based statistical modelling provides a robust, reproducible framework for identifying disease-associated gene expression changes (Ritchie *et al.*, 2015). In this study, three independently generated microarray datasets from NCBI GEO — GSE9150 (dust allergy), GSE23552 (CRSwNP), and GSE65204 (allergic asthma) — were processed using this pipeline to generate disease-specific differential expression profiles for cross-disease comparison.

1.4 FUNCTIONAL ENRICHMENT ANALYSIS AND CONSENSUS GENE DERIVATION

Functional enrichment analysis converts differentially expressed gene lists into interpretable biological information by testing whether genes are statistically enriched within predefined biological gene sets from GO, KEGG, Reactome, and other databases (Reimand *et al.*, 2019). Three complementary tools — Metascape, g:Profiler, and Enrichr — were applied to the shared gene set to maximise biological coverage and minimise platform-specific bias (Zhou *et al.*, 2019; Raudvere *et al.*, 2019; Kuleshov *et al.*, 2016). Integration of significance scores from Metascape and g:Profiler through negative log transformation of adjusted p-values yielded nine biologically consensus genes — CCL5, CCL20, IL7R, IDO1, CD36, CASP4, CTSC, CORO1A, and PAG1 — representing the most robustly enriched candidates across multiple independent analytical platforms.

1.5 PROTEIN–PROTEIN INTERACTION NETWORK ANALYSIS AND HUB GENE IDENTIFICATION

Network-based biology identifies hub genes — highly connected regulatory nodes — which exert disproportionate regulatory influence within biological interaction networks and frequently correspond to disease drivers or therapeutic targets (Gysi *et al.* 2021). The STRING database was used to construct a PPI network of the nine consensus genes at a medium confidence threshold (score > 0.4), which was imported into Cytoscape for visualisation and topological analysis (Szkłarczyk *et al.*, 2019; Otasek *et al.* 2019). Hub gene identification was performed using the CytoHubba plugin applying four centrality measures simultaneously — Maximal Clique Centrality (MCC), Degree, Closeness, and Betweenness — ensuring that hub gene designations are not artefacts of any single metric (Chin *et al.*, 2014; Tang *et al.* 2015). Genes ranking consistently at the top across all four measures were designated as hub genes, resulting in the identification of CCL5, CCL20, IL7R, and IDO1.

1.6 MULTI-OMICS INTEGRATION FOR BIOLOGICAL VALIDATION

Single-omics analyses are inherently limited because mRNA levels explain only 40–60% of protein abundance variance, with the remainder attributable to post-transcriptional regulation, translational efficiency, protein stability, and metabolic feedback (Chen *et al.*, 2023). To confirm that the shared transcriptional signatures are functionally active at the protein and metabolite levels, this study integrates transcriptomic, proteomic, and metabolomic data from sinusitis (CRSwNP) as a representative disease, selected based on the availability of published complementary datasets.

Pathway enrichment analysis was performed independently for each omics layer — using Metascape for transcriptomics and proteomics, and MetaboAnalyst for metabolomics — and results were integrated using a Python-based quantitative cross-omics framework. For each candidate pathway, a composite convergence score was calculated as: Composite Score = $-\log_{10}(\text{p-value}) \times \text{Combined Score} \times \text{Consistency Score} \times \text{Overlap Percentage}$. Pathways supported across two or

more omics layers were classified as convergent. A pan-grouped analysis then aggregated individual pathways into broader biological themes, and a bridge step compared pan-transcriptomic pathway signatures with sinusitis multi-omics results to confirm cross-study consistency (Subramanian *et al.*, 2020; Agache *et al.*, 2025).

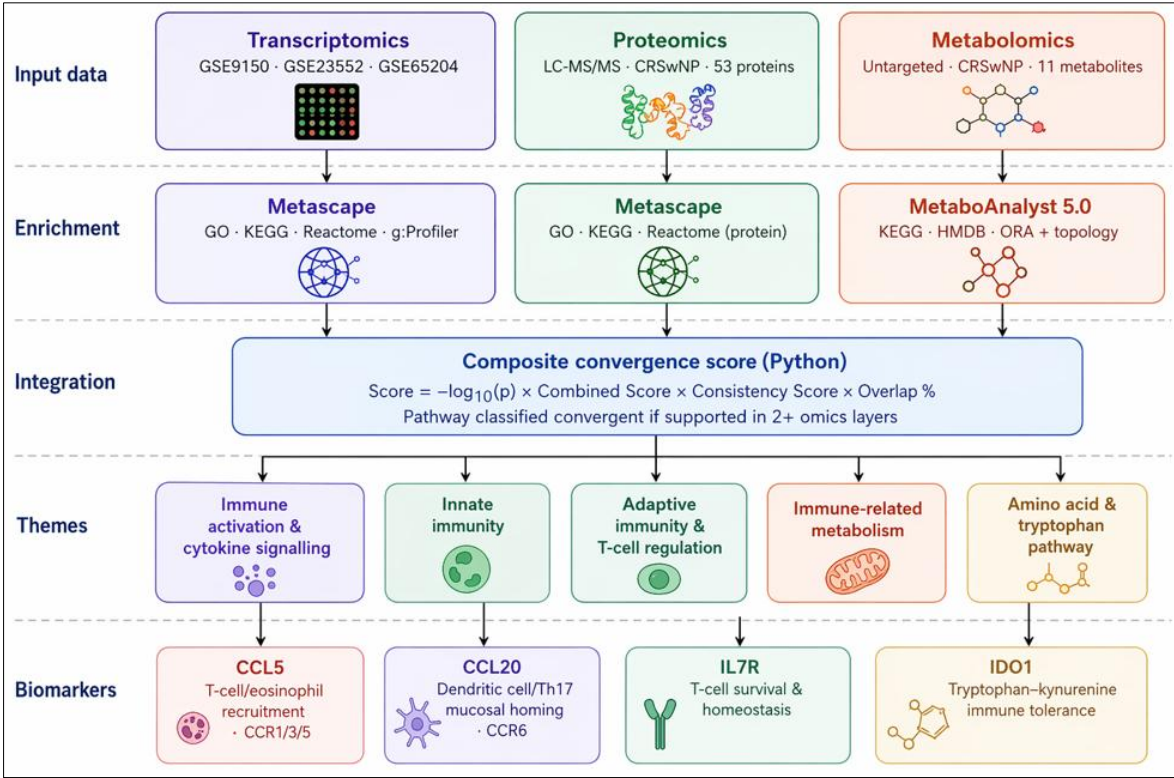


Fig: 1.3: Multi-omics Integration Framework

Multi-omics integration framework for cross-omics pathway convergence in CRSwNP (Zhou *et al.*, 2019; Raudvere *et al.*, 2019; Pang *et al.*, 2021; Subramanian *et al.*, 2020).

1.7 CANDIDATE BIOMARKERS AND MACHINE LEARNING EVALUATION

Four candidate biomarkers — CCL5, CCL20, IL7R, and IDO1 — were prioritised based on consistent top ranking across all four CytoHubba centrality measures, robust enrichment across all three functional annotation platforms, and support from more than one omics layer in the sinusitis multi-omics analysis. These candidates collectively span cytokine-mediated immune cell recruitment (CCL5, CCL20), T-cell homeostatic regulation (IL7R), and immunometabolic

reprogramming through the tryptophan–kynurenine pathway (IDO1), each supported by recent experimental and clinical literature (Dorismond *et al.*, 2025; Li *et al.*, 2024; Le Mai *et al.*, 2020; Miao *et al.*, 2024).

Machine learning evaluation was performed using Random Forest (RF) and Logistic Regression (LR) classifiers on the GSE65204 dataset with five-fold cross-validation. A two-part design was employed: Part 1 used all gene expression features to characterise the genome-wide discriminative landscape, while Part 2 used only the four candidate biomarkers to directly assess their discriminative capacity for distinguishing disease from healthy states (Peng *et al.*, 2021). The findings of this study contribute a rigorous, multi-evidence computational framework for pan-disease biomarker discovery that is fully reproducible and scalable to other inflammatory disease contexts.

CHAPTER 2

LITERATURE SURVEY

2.1 PATHOPHYSIOLOGY OF ALLERGIC RESPIRATORY DISEASES

2.1.1 *The United Airways Concept*

The united airways model proposes that the upper and lower respiratory tracts form a biologically and immunologically unified system, where inflammation in one compartment amplifies disease in anatomically distant regions (Bousquet *et al.*, 2019). This concept is validated by the high co-occurrence rates of allergic rhinitis, CRSwNP, and allergic asthma, their shared allergen triggers, overlapping immune endotypes, and convergent therapeutic responses (Iyer *et al.*, 2024). Systematic cross-disease molecular investigation, as performed in the present study, is a principled response to the limitations of disease-siloed research (Shin *et al.*, 2023).

2.1.2 *Immunopathogenesis of HDM Allergy*

House dust mite (HDM) allergy is driven by sensitisation to *Dermatophagoides pteronyssinus* and *Dermatophagoides farinae* allergens. The principal allergen Der p 1 is a cysteine protease that disrupts nasal epithelial barrier integrity through proteolytic cleavage of occludin and claudin-1 (Huang *et al.*, 2023). In sensitised individuals, IgE-mediated mast cell and basophil degranulation releases histamine, leukotrienes, and pro-inflammatory cytokines including IL-4, IL-5, IL-6, and IL-8, establishing a cytokine milieu that sustains chronic eosinophilic inflammation and mucosal remodelling (Kicic *et al.*, 2025).

2.1.3 *Molecular Pathogenesis of CRSwNP*

CRSwNP is characterised by a type 2 immune-dominant inflammatory phenotype driven by epithelial alarm cytokines, Th2-polarised T cells, and local IgE class switching (Ha and Kim, 2025). Core pathological features include eosinophilic infiltration, elevated local IgE, submucosal oedema, and epithelial barrier dysfunction — all promoted by IL-4 and IL-13 signalling through shared IL-4R α receptor subunits and JAK-STAT6 cascades (Chiang and Lee, 2023). The

sinonasal epithelium functions as an active immunological interface that senses microbial and allergen stimuli through pattern recognition receptors and orchestrates immune cell recruitment (Huang and Cho, 2023).

2.1.4 Immunological Mechanisms of Allergic Asthma

Allergic asthma is defined by reversible airflow obstruction, bronchial hyperresponsiveness, mucus hypersecretion, and progressive airway remodelling (Iyer *et al.*, 2024). The dominant immunological driver is the type 2 cytokine environment established by Th2 cells and ILC2s, with IL-5 promoting eosinophil recruitment, IL-4 and IL-13 driving mucus production and IgE class switching, and upstream epithelial alarmins TSLP, IL-25, and IL-33 initiating the cascade (Ha and Kim, 2025). The frequent co-occurrence of asthma with CRSwNP in up to 80% of patients further supports the united airways model (Bousquet *et al.*, 2019).

2.2 TRANSCRIPTOMIC APPROACHES IN ALLERGIC DISEASE RESEARCH

2.2.1 Microarray Technology and GEO Repositories

Transcriptomics enables genome-wide, simultaneous measurement of RNA expression levels across thousands of genes, enabling molecular phenotyping of diseased versus healthy tissues. The NCBI Gene Expression Omnibus (GEO) archives over 225,000 publicly available functional genomics datasets, providing open access to microarray and RNA-sequencing data from diverse disease contexts (Clough & Barrett 2016). The systematic reanalysis of publicly deposited datasets from independent cohorts and disease contexts is a scientifically rigorous approach to identifying molecular signals that are robust and generalisable rather than study-specific (Rosati *et al.*, 2024).

2.2.2 Differential Expression Analysis

Standard microarray differential expression analysis encompasses background correction, RMA or quantile normalisation, PCA quality control, and statistical modelling using the limma framework with empirical Bayes variance moderation (Ritchie *et al.*, 2015). Limma's shrinkage of gene-wise variance estimates toward a global prior improves sensitivity, particularly in small-sample studies common in

publicly deposited GEO datasets. To minimise cross-dataset batch confounding, differential expression analysis was performed independently within each dataset in this study, ensuring disease-control comparisons occur within a homogeneous technical context (Luo *et al.* (2015)).

2.3 FUNCTIONAL ENRICHMENT ANALYSIS TOOLS

2.3.1 *Metascape*

Metascape integrates GO biological processes, KEGG, Reactome, WikiPathways, and protein interaction data into a single analytical platform, applying network-based pathway clustering to reduce enrichment redundancy and highlight dominant biological themes (Zhou *et al.*, 2019). Its log-transformed p-value (log B) scoring provides a quantitative metric suitable for cross-tool significance comparison.

2.3.2 *g:Profiler*

g:Profiler performs multi-source functional enrichment across GO, KEGG, Reactome, WikiPathways, and other databases, applying the g:SCS algorithm for multiple testing correction. The g:SCS approach accounts for the non-independence of overlapping gene sets in biological databases — a known limitation of standard Benjamini-Hochberg correction — reducing both false positive and false negative enrichment calls (Raudvere *et al.*, 2019).

2.3.3 *Enrichr*

Enrichr aggregates data from a broad collection of curated gene set libraries including GO, KEGG, Reactome, WikiPathways, DisGeNET disease associations, and drug perturbation signatures, enabling functional associations not captured by pathway-focused tools alone (Kuleshov *et al.*, 2016). Using three enrichment platforms with distinct but overlapping database coverage substantially increases the robustness of the final consensus gene set relative to any single-tool analysis.

2.4 NETWORK BIOLOGY AND HUB GENE IDENTIFICATION

2.4.1 STRING Database

The STRING database is the most comprehensively curated resource for PPI network construction, integrating experimentally validated physical interactions, computationally predicted associations, text-mined interactions, and curated pathway data, assigning composite confidence scores to each interaction (Szklarczyk *et al.*, 2019). A medium confidence threshold of 0.4 is standard practice for disease gene network analysis, balancing network connectivity with interaction reliability (Li *et al.*, 2022).

2.4.2 Cytoscape and CytoHubba

Cytoscape is an open-source platform for molecular interaction network visualisation and analysis (Otasek *et al.* 2019). The CytoHubba plugin provides eleven topological scoring algorithms for ranking nodes within biological networks (Chin *et al.*, 2014; Tang *et al.* 2015). Maximal Clique Centrality (MCC), validated as the most accurate method for identifying essential proteins from PPI networks, is considered the gold standard within CytoHubba. Degree, Closeness, and Betweenness centrality capture direct connectivity, network communication efficiency, and regulatory bottleneck status respectively (). Combining all four measures ensures hub gene designations are not artefacts of a single algorithm.

2.5 PROTEOMICS AND METABOLOMICS IN CRSWNP

2.5.1 Proteomic Investigation

Label-free quantitative LC-MS/MS is the dominant quantitative proteomics technology for clinical biomarker research, enabling simultaneous identification and relative quantification of hundreds to thousands of proteins from biological fluids such as nasal lavage fluid (Hristova and Chan, 2019). The divergence between mRNA-level transcriptomic and protein-level proteomic observations — arising from post-transcriptional silencing, translational efficiency, and protein stability differences — means that proteomic data provide an independent

validation line that is not redundant with transcriptomic information (Chen *et al.*, 2023).

2.5.2 Metabolomic Profiling

Metabolomics occupies the most downstream position in the omics hierarchy, capturing the net biochemical output of all upstream genomic, transcriptomic, and proteomic regulatory programmes (Wishart, 2016). Untargeted metabolomic profiling of CRSwNP has identified disease-specific signatures involving tryptophan, phenylalanine, amino acid metabolism, and nitrogen metabolism — changes that directly connect to the IDO1 hub gene identified in pan-transcriptomic analysis (Miao *et al.*, 2024). Metabolomic pathway analysis was performed using MetaboAnalyst 5.0, which conducts over-representation analysis and pathway topology analysis against KEGG and HMDB databases (Pang *et al.*, 2021).

2.6 MULTI-OMICS INTEGRATION FRAMEWORKS

Multi-omics integration is motivated by the well-documented imperfect correlation between mRNA abundance and protein expression, which accounts for only 40–60% of protein level variance (Chen *et al.*, 2023). Pathway-level integration is the most tractable strategy for combining transcriptomic, proteomic, and metabolomic data from different experimental sources, as it does not require the same molecules to be directly measured across all platforms (Subramanian *et al.*, 2020). The Reactome pathway database provides a hierarchically organised, peer-reviewed framework for contextualising convergent findings within curated biological pathways, enabling precise identification of which molecular steps are supported by the experimental data (Gillespie *et al.*, 2022). Integrative omics has been endorsed by EAACI and other international allergy societies as the scientific standard for mechanistic disease understanding and biomarker discovery (Agache *et al.*, 2025; Jutel *et al.*, 2023).

2.7 LITERATURE EVIDENCE FOR CANDIDATE BIOMARKERS

2.7.1 CCL5

CCL5 (RANTES) is a CC-family chemokine that mediates recruitment of T lymphocytes, eosinophils, monocytes, and NK cells to sites of inflammation through CCR1, CCR3, and CCR5 receptors. Recent clinical evidence demonstrates that elevated CCL5 levels are statistically associated with distinct inflammatory endotypes in CRS, and that CCL5 modulates cytokine response networks in a phenotype-determining manner (Dorismond *et al.*, 2025).

2.7.2 CCL20

CCL20 (MIP-3 α), the sole ligand for CCR6, plays a non-redundant role in directing dendritic cells, immature B cells, and Th17 cells to mucosal surfaces. Experimental evidence confirms that CCL20–CCR6 signalling contributes directly to immune cell recruitment and amplification of inflammatory responses in allergic airway conditions (Li *et al.*, 2024).

2.7.3 IL7R

IL7R (CD127) encodes the alpha subunit of the IL-7 receptor, essential for T and B lymphocyte development and homeostatic T cell survival. Pharmacological blockade of IL7R signalling has been demonstrated to significantly reduce Th2 cytokine production — including IL-4, IL-5, and IL-13 — and eosinophilic airway infiltration in experimental allergic airway inflammation models (Le Mai *et al.*, 2020).

2.7.4 IDO1

IDO1 is the rate-limiting enzyme of the kynurenine pathway of tryptophan catabolism, strongly induced by IFN- γ . Its products modulate immune tolerance through tryptophan depletion-mediated GCN2 kinase activation in T effector cells and kynurenine-driven regulatory T cell differentiation via AhR activation (Miao *et al.*, 2024). The detection of tryptophan-related pathway enrichment in the

CRSwNP metabolome provides direct cross-omics mechanistic support for IDO1 as a candidate biomarker.

2.8 MACHINE LEARNING IN BIOMARKER DISCOVERY

2.8.1 Random Forest Classification

Random Forest is an ensemble algorithm that constructs multiple decision trees on bootstrap-sampled data, using random feature subsets at each node, and aggregates predictions through majority voting. Its resistance to overfitting and generation of feature importance scores make it one of the most widely adopted algorithms in transcriptomic biomarker discovery (Breiman, 2001; Peng *et al.*, 2021).

2.8.2 Logistic Regression and Cross-Validation

Logistic Regression is a parametric linear classifier that models log-odds of class membership, providing interpretable feature coefficients and a complementary linear perspective to the non-linear Random Forest (Peng *et al.*, 2021). Five-fold cross-validation — the standard technique for producing unbiased performance estimates from limited data — was applied throughout, generating accuracy, AUC-ROC, precision, recall, and F1 score estimates representative of generalised classification performance ().

2.8.3 External Dataset Validation

External dataset validation tests whether computationally derived candidate biomarkers show significant differential expression in independent datasets not used during the discovery phase. Variable statistical significance across independent cohorts reflects known biological heterogeneity — differences in disease endotype, tissue type, severity distribution, and study population — and is expected rather than indicative of biomarker invalidity (Rosati *et al.*, 2024).

CHAPTER 3

AIM AND SCOPE

3.1 AIM

The aim of this study is to identify shared immune and inflammatory molecular pathways across dust allergy, chronic rhinosinusitis with nasal polyps (CRSwNP), and allergic asthma using pan-transcriptomic analysis, and to validate their functional relevance through multi-omics integration at the protein and metabolite levels. The study further aims to prioritise biologically credible candidate biomarkers based on cross-disease consistency, network importance, and multi-omics support.

3.2 OBJECTIVES

The following objectives were formulated to achieve the study aim through a sequential analytical pipeline.

- Three publicly available microarray gene expression datasets were retrieved from the NCBI Gene Expression Omnibus (GEO) — GSE9150 representing dust allergy, GSE23552 representing chronic rhinosinusitis with nasal polyps, and GSE65204 representing allergic asthma. These datasets were selected on the basis of clearly defined disease and healthy control sample groups, use of nasal or airway epithelial tissue relevant to the united airways disease continuum, and availability of platform annotation files. A standardised R-based pipeline was applied to each dataset independently, covering automated data download using the GEOquery package, background correction via the Robust Multi-array Average (RMA) method, quantile normalisation to remove inter-array technical variation, and principal component analysis (PCA)-based quality control to identify outlier samples prior to downstream analysis. Differential gene expression analysis was subsequently performed using the limma package with empirical Bayes moderation, contrasting disease samples against healthy controls. Differentially expressed gene (DEG) lists were generated at two thresholds: a discovery threshold ($p < 0.05$, $|\log_2FC| >$

0.5) and a strict high-confidence threshold (Benjamini–Hochberg adjusted $p < 0.05$, $|\log_2FC| \geq 1.0$). Microarray probe identifiers were mapped to HGNC gene symbols using vendor-supplied platform annotation files, and where multiple probes mapped to the same gene, the probe with the highest mean expression was retained. Cross-disease gene overlap was then identified by computing set intersections in R at both thresholds, retaining only genes differentially expressed across all three diseases simultaneously, yielding a 15-gene discovery set and a 4-gene strict set. A biological sanity check confirmed directional consistency and functional coherence of the overlapping gene sets before proceeding to enrichment analysis.

- The 15-gene discovery set was submitted to three independent functional enrichment platforms — Metascape, g:Profiler, and Enrichr — to identify the biological processes and pathways most significantly associated with the shared gene signature. Metascape integrated GO biological processes, KEGG, Reactome, and WikiPathways with network-based clustering to reduce redundancy among enriched terms, using the log B value as the primary significance metric. g:Profiler applied the g:SCS multiple testing correction algorithm across the same database sources, providing a statistically conservative and non-redundant enrichment profile. Significance scores from both platforms were integrated through a semi-quantitative composite scoring approach, and genes achieving consistently high enrichment significance across both tools were designated as consensus genes, yielding nine biologically high-confidence candidates: CCL5, CCL20, IL7R, IDO1, CD36, CASP4, CTSC, CORO1A, and PAG1. Enrichr was applied as a third independent platform for cross-validation across GO Biological Process, Reactome, KEGG, DisGeNET, and OMIM libraries. A protein–protein interaction (PPI) network for the nine consensus genes was subsequently constructed using the STRING database (v11.5) at a medium confidence threshold (combined score ≥ 0.4) and imported into Cytoscape (v3.10) for visualisation using the Prefuse force-directed layout. Hub gene identification was performed using the CytoHubba plugin by applying four centrality measures simultaneously — Maximal Clique Centrality (MCC), Degree, Closeness, and Betweenness — and genes

ranking consistently at the top positions across all four metrics were designated as hub genes, yielding four final candidates: CCL5, CCL20, IL7R, and IDO1.

- Multi-omics validation was conducted using CRSwNP as the representative disease, selected on the basis of the availability of published complementary proteomic and metabolomic datasets. Three molecular layers were integrated: for the transcriptomic layer, upregulated differentially expressed genes from GSE23552 were submitted to Metascape for pathway enrichment analysis; for the proteomic layer, 53 significantly upregulated proteins from a published label-free LC-MS/MS study of CRSwNP nasal lavage fluid were submitted to Metascape; and for the metabolomic layer, 11 disease-associated metabolites from a published untargeted CRSwNP metabolomics study were submitted to MetaboAnalyst 5.0 for pathway enrichment and topology analysis. A Python-based quantitative cross-omics pathway integration framework was developed to combine enriched pathways from all three layers into a single composite convergence score, computed as: $\text{Composite Score} = -\log_{10}(\text{adjusted p-value}) \times \text{Combined Score} \times \text{Consistency Score} \times \text{Overlap Percentage}$. Pathways supported in two or more omics layers were classified as convergent. A pan-grouped analysis further aggregated individual pathways into five broader biological theme categories — immune activation and cytokine signalling, innate immunity, adaptive immunity, immune-related metabolism, and amino acid and nitrogen metabolism — and a bridge step directly compared pan-transcriptomic pathway signatures with sinusitis multi-omics findings to confirm cross-study molecular coherence. The four hub genes were contextualised using Reactome pathway visualisation, and their functional roles in allergic respiratory diseases were confirmed through a two-phase validation process: systematic literature review using peer-reviewed publications from PubMed, and independent external transcriptomic validation using three GEO datasets not used in the discovery phase — GSE179265, GSE75011, and GSE72713.

- The discriminative performance of the four finalised candidate biomarkers — CCL5, CCL20, IL7R, and IDO1 — was evaluated using supervised machine learning classifiers applied to the GSE65204 allergic asthma dataset, which was selected for its largest sample size ($n = 69$) among the three discovery datasets. Two classifiers were implemented in Python using scikit-learn: Random Forest (RF), an ensemble tree-based classifier resistant to overfitting in high-dimensional transcriptomic data, and Logistic Regression (LR), a linear parametric classifier providing interpretable feature coefficients. Both were evaluated under five-fold stratified cross-validation throughout. A two-part experimental design was applied: Part 1 trained both classifiers on all genome-wide gene expression features to characterise the overall discriminative landscape of the dataset and identify the top 15 most important genes by genome-wide criteria; Part 2 retrained both classifiers using only the four candidate biomarkers as input features to directly evaluate their classification capacity as a constrained panel. Classification performance was reported using accuracy, AUC-ROC, precision, recall, F1 score, and mean cross-validated accuracy. ROC curves and feature importance bar charts were generated for both parts, and the relative feature importance rankings from Part 2 were compared against the CytoHubba network centrality rankings to assess cross-evidence convergence between network topological prominence and statistical discriminative performance.

Collectively, these objectives test the central hypothesis of this study: that immune and inflammatory molecular pathways are shared across dust allergy, CRSwNP, and allergic asthma, and that these mechanisms are consistently supported across transcriptomic, proteomic, and metabolomic levels of biological analysis.

3.3 SCOPE

The scope of this study covers a fully computational, integrative investigation of shared molecular mechanisms across the three selected allergic respiratory diseases. The work encompasses pan-transcriptomic discovery using publicly available microarray datasets from the NCBI GEO database, cross-disease gene overlap analysis, multi-platform functional enrichment analysis, PPI network construction and hub gene identification, multi-omics validation using published proteomic and metabolomic data for CRSwNP as a representative disease, quantitative cross-omics pathway integration, biomarker finalisation with literature and external dataset validation, and machine learning-based biomarker performance evaluation. The study is entirely computational and does not involve experimental work, clinical sample collection, or wet laboratory analyses.

CHAPTER 4

MATERIALS AND METHODS

The complete analytical workflow is illustrated in Figure 4.1. The following sections describe each stage of the pipeline in detail.

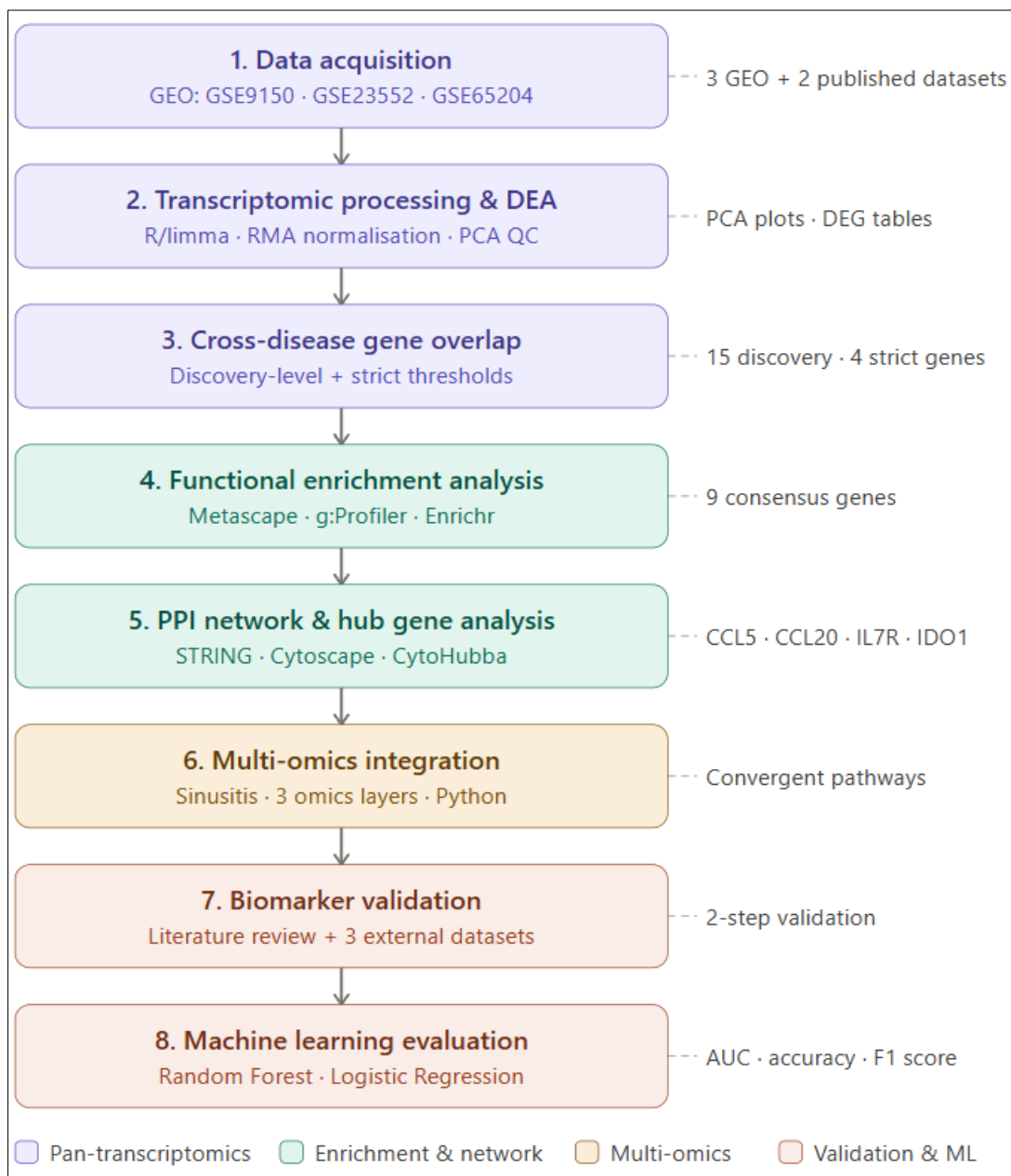


Fig:4.1: Overview of the eight-stage integrated analytical pipeline used in this study.

4.1 DATA SOURCES AND STUDY DESIGN

4.1.1 Transcriptomic Datasets

Three microarray transcriptomic datasets were retrieved from the NCBI Gene Expression Omnibus (GEO; Clough & Barrett 2016), each representing one of the three allergic respiratory diseases under investigation. Datasets were selected based on four criteria: clearly defined disease and healthy control sample groups; use of nasal or airway epithelial tissue relevant to the united airways disease continuum; adequate sample size for reliable differential expression analysis; and availability of platform annotation files. The selected datasets are summarised in Table 4.1.

Table 4.1: Summary of the three GEO microarray datasets used for pan-transcriptomic analysis.

Parameter	Dust Allergy	Sinusitis (CRSwNP)	Allergic Asthma
GEO Accession	GSE9150	GSE23552	GSE65204
Data Type	Microarray	Microarray (exon array)	Microarray
Total Samples	20	39	69
Tissue	Nasal epithelial cells	Nasal mucosa & nasal polyp tissue	Nasal epithelial cells
Role in Study	Discovery	Discovery + multi-omics validation	Discovery + machine learning

4.1.2 Proteomic and Metabolomic Datasets

Since the production of novel proteomic and metabolomic data falls outside the computational scope of this project, published peer-reviewed datasets were sourced from the scientific literature for the multi-omics validation stage. CRSwNP was selected as the representative disease because high-quality published

proteomic and metabolomic datasets are available for this condition. The proteomic dataset was derived from a label-free quantitative LC-MS/MS study of CRSwNP nasal lavage fluid, yielding 230 identified proteins of which 53 were significantly upregulated (fold change > 1.5, $p < 0.05$). The metabolomic dataset comprised 11 disease-associated metabolites from a published untargeted metabolomics study of CRSwNP samples. Both datasets were pre-processed and cleaned as described in Section 4.4.

4.2 TRANSCRIPTOMIC DATA PROCESSING AND DIFFERENTIAL EXPRESSION ANALYSIS

4.2.1 Data Download and Preprocessing

All three transcriptomic datasets were programmatically downloaded from NCBI GEO using the GEOquery R package (Clough & Barrett (2016)). A single universal R script was written to process all datasets consistently. Three preprocessing steps were applied to each dataset: (i) background correction using the Robust Multi-array Average (RMA) method, to remove non-specific hybridisation signal and normalise within-array probe intensities; (ii) quantile normalisation across all arrays within each dataset, to correct systematic inter-array signal variation; and (iii) PCA using the `prcomp()` function in R, to confirm expected sample groupings and detect any outlier samples prior to downstream analysis. PCA quality control plots for all three datasets are shown in Chapter 5.

4.2.2 Probe-to-Gene Annotation

Platform-specific probe identifiers were mapped to HGNC gene symbols using annotation files downloaded from the corresponding GPL platform records on GEO. Probe-level expression values were converted to gene-level values. Where multiple probes mapped to the same gene symbol, the probe with the highest mean expression across all samples was retained as the representative gene-level measurement, following the established standard procedure (Ritchie *et al.*, 2015).

4.2.3 Differential Expression Analysis

Differential gene expression analysis was performed independently for each dataset using the limma package in R/Bioconductor (Ritchie *et al.*, 2015). A linear model was fitted to contrast disease samples against healthy controls, and variance estimation was moderated using empirical Bayes moderation through the eBayes() function, which improves statistical stability and sensitivity in datasets with small sample sizes. Two significance thresholds were applied: a discovery threshold ($p < 0.05$, $|\log_2FC| > 0.5$) and a strict high-confidence threshold (Benjamini–Hochberg adjusted $p < 0.05$, $|\log_2FC| \geq 1.0$). Both DEG sets were carried forward to the cross-disease overlap analysis.

4.3 CROSS-DISEASE GENE OVERLAP ANALYSIS

4.3.1 Gene Set Derivation

Following independent differential expression analyses of all three datasets, cross-disease gene overlap was identified using set intersection operations in R. The intersection was computed separately for both significance thresholds, retaining only genes present in the DEG lists of all three diseases simultaneously. This produced two cross-disease gene sets: a 15-gene discovery set of genes showing nominally significant differential expression across all three diseases, and a 4-gene strict set of genes meeting the high-confidence threshold in all three diseases. Venn diagrams for both overlaps are presented in Chapter 5.

4.3.2 Biological Sanity Check

A sanity check was performed on the overlapping gene sets to verify that the identified genes showed consistent direction of expression change across all three diseases and were functionally coherent in terms of involvement in immune and inflammatory processes. This step confirmed the suitability of the gene sets for downstream functional enrichment analysis.

4.4 FUNCTIONAL ENRICHMENT ANALYSIS

4.4.1 Metascape

The 15-gene discovery set was submitted to Metascape (Zhou *et al.*, 2019) for functional enrichment analysis. Metascape integrates GO biological processes, KEGG, Reactome, and WikiPathways databases, and applies network-based clustering to reduce redundancy among enriched terms. Homo sapiens was selected as the organism with the whole genome as the background reference. The log B value was used as the primary significance metric. Enriched GO biological process bar charts are shown in Chapter 5.

4.4.2 g:Profiler

The same 15-gene list was submitted to g:Profiler (Raudvere *et al.*, 2019) for independent enrichment analysis across GO biological processes, KEGG, Reactome, and WikiPathways. g:Profiler applies the g:SCS algorithm for multiple testing correction, which accounts for the non-independence of overlapping gene sets in biological databases — an advantage over standard Benjamini–Hochberg correction. Adjusted p-values from g:Profiler were used alongside Metascape log B values for consensus gene derivation.

4.4.3 Consensus Gene Derivation

Enrichment output tables from Metascape and g:Profiler were compared through a semi-quantitative integration approach. Genes showing consistently significant enrichment across both platforms with functionally coherent annotations were retained. This produced nine consensus genes: CCL5, CCL20, IL7R, IDO1, CD36, CASP4, CTSC, CORO1A, and PAG1.

4.4.4 Enrichr Validation

The nine-gene consensus set was submitted to Enrichr (Kuleshov *et al.*, 2016) for independent cross-platform validation. Enrichr was queried across GO Biological Process, Reactome, KEGG, DisGeNET, and OMIM libraries to confirm the biological themes identified through Metascape and g:Profiler and to establish disease-specific associations of the consensus genes with allergic and inflammatory conditions.

4.5 PROTEIN-PROTEIN INTERACTION NETWORK CONSTRUCTION AND HUB GENE IDENTIFICATION

4.5.1 STRING Network Construction

A PPI network for the nine consensus genes was constructed using the STRING database (v11.5; Szklarczyk *et al.*, 2019), which integrates experimentally validated physical interactions, predicted interactions, text-mined co-mentions, and curated database interactions. Homo sapiens was selected as the organism. A medium confidence threshold (combined score ≥ 0.4) was applied, which follows standard practice in published inflammatory disease gene network studies.

4.5.2 Cytoscape Network Visualisation

The STRING-derived PPI network was exported to Cytoscape (v3.10; Otasek *et al.* 2019) using the STRING API. The network was visualised using the Prefuse force-directed layout, with edge thickness proportional to interaction confidence scores.

4.5.3 Hub Gene Identification Using CytoHubba

Hub gene identification was performed using the CytoHubba plugin (Chin *et al.*, 2014; Tang *et al.* 2015) by applying four centrality measures simultaneously to rank all nodes in the network:

Maximal Clique Centrality (MCC): Scores nodes by membership in maximal cliques. MCC is validated as the gold standard measure in CytoHubba for identifying essential protein hubs.

Degree Centrality: Counts direct interaction partners for each node, reflecting its immediate connectivity in the network.

Closeness Centrality: Measures the mean shortest path distance from a node to all others, identifying genes with the greatest potential to propagate signals across the network.

Betweenness Centrality: Counts shortest paths between all node pairs that pass through a given node, identifying regulatory bottlenecks controlling cross-module communication.

Genes ranking consistently at the top positions across all four measures were designated as hub genes, ensuring the designations are not artefacts of any single metric. Four hub genes were identified: CCL5, CCL20, IL7R, and IDO1.

4.6 MULTI-OMICS DATA INTEGRATION FOR SINUSITIS

4.6.1 Rationale for Selecting CRSwNP as the Representative Disease

CRSwNP was selected as the model disease for multi-omics validation because high-quality published proteomic and metabolomic datasets are available specifically for this condition, and because CRSwNP is the most extensively studied immunopathological condition among the three allergic respiratory diseases investigated in this study.

4.6.2 Transcriptomic Layer

The GSE23552 dataset processed in Section 4.2 was reused for sinusitis-specific transcriptomic pathway analysis. Upregulated genes ($\log_2FC > 0$, $p < 0.05$) from the CRSwNP differential expression analysis were submitted to Metascape for pathway enrichment analysis using the same parameters described in Section 4.4.1. Enriched pathways were exported for downstream cross-omics integration.

4.6.3 Proteomic Layer

The published LC-MS/MS CRSwNP nasal lavage fluid proteomic dataset was pre-processed by removing entries with missing protein identifiers or quantification values, harmonising protein names to HGNC gene symbols, and excluding non-human contaminant proteins. From the 230 filtered proteins, 53 significantly upregulated proteins (fold change > 1.5, $p < 0.05$) were submitted to Metascape for pathway enrichment analysis, enabling direct pathway-level comparison with the transcriptomic enrichment results.

4.6.4 Metabolomic Layer

The 11 disease-associated metabolites from the published CRSwNP untargeted metabolomics study were submitted to MetaboAnalyst 5.0 (Pang *et al.*, 2021) for pathway enrichment and topology analysis. Both upregulated and downregulated metabolites were included. Pathway topology analysis calculated the importance score of each metabolite based on its network centrality. Enriched metabolic pathways were exported for cross-omics integration.

4.7 QUANTITATIVE CROSS - OMICS PATHWAY INTEGRATION

4.7.1 Composite Score Computation

A custom Python script was developed to integrate enriched pathways from all three omics layers into a single quantitative measure of convergence. For each candidate pathway, a composite score was calculated using the following formula:

$$\text{Composite Score} = -\log_{10}(\text{adjusted p-value}) \times \text{Combined Score} \times \text{Consistency Score} \times \text{Overlap Percentage}$$

The $-\log_{10}(\text{adjusted p-value})$ reflects enrichment significance; Combined Score accounts for effect magnitude and biological support; Consistency Score reflects the number of omics layers supporting the pathway; and Overlap Percentage reflects the proportion of pathway members observed experimentally. The multiplicative structure ensures that a pathway must perform well across all

components simultaneously to achieve a high score. Pathways supported in two or more omics layers were classified as convergent.

4.7.2 Pan-Grouped Analysis and Pathway Agreement

Individual enriched pathways were aggregated into five broader biological theme categories through a pan-grouped analysis: (i) immune activation and cytokine signalling; (ii) innate immunity; (iii) adaptive immunity; (iv) immune-related metabolism; and (v) amino acid and nitrogen metabolism. A pathway agreement analysis then identified which themes were independently detected in two or more omics layers, representing the most cross-validated findings of the integration framework.

4.7.3 Bridge Step: Pan-Transcriptomic Versus Multi-Omics Comparison

A bridge analysis was performed to compare biological themes from the pan-transcriptomic analysis (all three diseases) with those from the sinusitis multi-omics integration. Themes found to be concordant between both analyses — including immune activation, cytokine signalling, inflammatory response, leukocyte migration, innate and adaptive immunity, and immune-related metabolism — were considered the most robustly supported findings. Concordant themes are visualised using Python-generated comparison bar graphs in Chapter 5.

4.7.4 Reactome Pathway Contextualisation

The four candidate biomarker genes were submitted to the Reactome pathway database (Gillespie *et al.*, 2022) for pathway enrichment and hierarchical pathway visualisation. Reactome's reaction-based pathway modelling was used to precisely identify the cytokine signalling and interleukin signalling components enriched by the candidate biomarkers, providing mechanistic depth to the multi-omics integration findings.

4.8 BIOMARKER FINALISATION AND VALIDATION

4.8.1 Biomarker Prioritisation Criteria

The final set of candidate biomarkers was selected from the four hub genes identified in Section 4.5.3 based on three converging lines of evidence: (i) consistent top-ranking across all four CytoHubba centrality metrics; (ii) robust enrichment significance across all three functional annotation platforms; and (iii) pathway-level support in at least two of the three omics layers in the sinusitis multi-omics framework. Based on these criteria, CCL5, CCL20, IL7R, and IDO1 were finalised as the candidate pan-disease biomarkers.

4.8.2 Literature Validation

A two-tiered systematic literature validation was conducted for each of the four finalised biomarkers. Peer-reviewed publications from PubMed were reviewed to confirm experimental, clinical, mechanistic evidence for the role of each biomarker in allergic respiratory diseases. Key aspects examined included immune cell recruitment, cytokine signalling regulation, T-cell activity, and inflammatory immunometabolism. Specific findings for each biomarker are discussed in Chapter 5.

4.8.3 External Dataset Validation

Independent external validation was performed using three GEO transcriptomic datasets not used in the primary discovery phase: GSE179265, GSE75011, and GSE72713. For each external dataset, differential expression of the four candidate biomarkers was assessed. A biomarker was considered supported if it showed statistically significant differential expression (adjusted $p < 0.05$) in a direction consistent with the discovery results. Validation results across all three datasets are tabulated in Chapter 5.

4.9 MACHINE LEARNING-BASED BIOMARKER PERFORMANCE EVALUATION

4.9.1 Dataset Selection and Rationale

Machine learning evaluation was performed using the GSE65204 allergic asthma dataset (69 samples), which was also used in the pan-transcriptomic discovery phase. This dataset was selected because it has the largest sample size among the three datasets — a critical consideration for training reliable supervised classifiers — and provides clearly defined binary class labels. The limitation of using a discovery-phase dataset for machine learning evaluation is acknowledged and is discussed in the Discussion section of Chapter 5.

4.9.2 Classifiers and Cross-Validation

Two supervised classifiers were applied: Random Forest (RF; Breiman, 2001), an ensemble tree-based classifier resistant to overfitting in high-dimensional data, and Logistic Regression (LR; Peng *et al.*, 2021), a linear parametric classifier providing interpretable feature coefficients. Both were implemented in Python using scikit-learn with five-fold stratified cross-validation throughout. A two-part experimental design was applied: Part 1 trained both classifiers on all genome-wide gene expression features to establish a discriminative performance baseline; Part 2 retrained both classifiers using only the four finalised biomarkers as input, directly evaluating their classification capacity as a constrained panel. Performance was assessed using accuracy, AUC-ROC, precision, recall, F1 score, and mean cross-validated accuracy. ROC curves and feature importance bar charts were generated for visual analysis.

4.9.3 Feature Importance Interpretation

For Part 1, the 15 most important gene features from the Random Forest model were identified and discussed in relation to the four selected biomarkers. The

absence of biomarker genes from the Part 1 top-15 list is attributed to the dataset-specific biology of the GSE65204 paediatric atopic asthma cohort, where gene-level importance scores reflect cohort-specific expression patterns rather than pan-disease shared mechanisms. For Part 2, the individual importance scores of the four biomarkers were ranked to assess their relative contributions to biomarker panel classification performance.

All datasets used in this study were publicly available through NCBI GEO and peer-reviewed published literature. No human participants were recruited, no patient data were directly accessed, and no ethical approval was required for this study.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 TRANSCRIPTOMIC DATA PRE-PROCESSING AND QUALITY CONTROL

Three publicly available microarray gene expression datasets were retrieved from the NCBI Gene Expression Omnibus (GEO) repository (Clough & Barrett 2016) for pan-transcriptomic analysis. The datasets represented three distinct allergic respiratory conditions: dust allergy (GSE9150), chronic rhinosinusitis with nasal polyps (GSE23552), and allergic asthma (GSE65204). Datasets were selected based on the availability of clearly demarcated disease and healthy control sample groups, use of nasal or airway epithelial tissue relevant to the united airways disease continuum, and use of established Affymetrix or equivalent microarray platforms with available platform annotation files.

A customised R programming pipeline was developed to perform the following pre-processing steps for each dataset independently: (i) automated data download using the GEOquery package (Clough & Barrett (2016)); (ii) background correction using the Robust Multi-array Average (RMA) algorithm; (iii) quantile normalisation to remove inter-array technical variation; (iv) principal component analysis (PCA)-based quality control to identify and exclude outlier samples; and (v) differential expression analysis using the limma framework with empirical Bayes moderation (Ritchie *et al.*, 2015). Probe-level intensity values were subsequently mapped to HGNC gene symbols using platform-specific annotation files retrieved for each GPL platform.

PCA quality control confirmed clear separation of disease and control sample groups along the first principal component in all three datasets — GSE9150, GSE23552, and GSE65204 — validating the normalised expression data for downstream differential expression analysis. No outlier samples requiring exclusion were identified in any dataset.

5.2 CROSS-DISEASE GENE OVERLAP ANALYSIS

Differential expression analysis was performed independently for each dataset using the limma package, comparing disease samples against healthy controls. Two significance thresholds were applied: a discovery threshold ($p < 0.05$, $|\log_2FC| > 0.5$) and a strict high-confidence threshold (Benjamini–Hochberg adjusted $p < 0.05$, $|\log_2FC| \geq 1.0$).

At the discovery threshold, 15 genes were identified as differentially expressed across all three allergic respiratory diseases (GSE9150, GSE23552, and GSE65204). Figure 5.1 displays the Venn diagram of gene overlap across the three datasets.

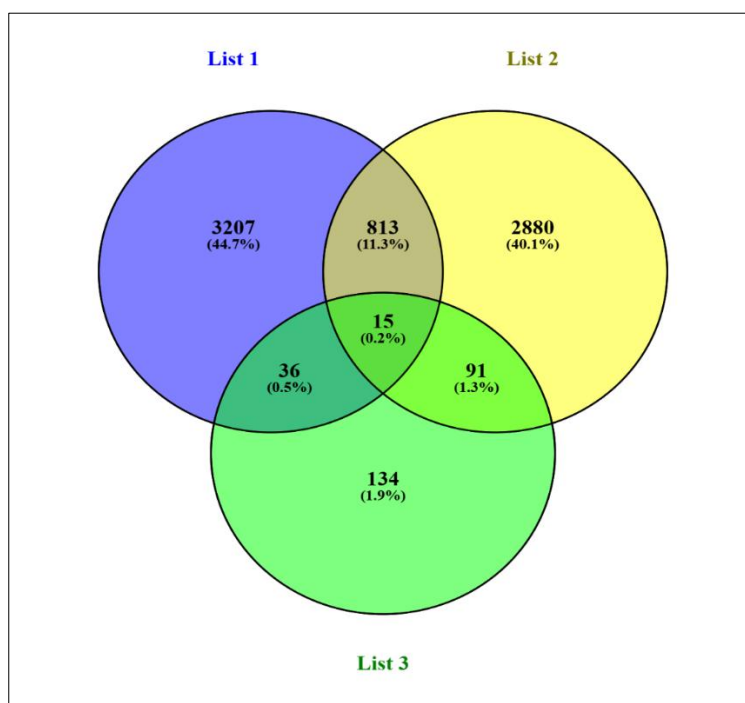


Fig:5.1: Venn diagram showing 15 discovery genes shared across dust allergy, CRSwNP, and allergic asthma

The 15 shared discovery genes span functionally diverse categories encompassing immune cell recruitment (CCL5, CCL20), T-cell regulation (IL7R, PAG1), innate immune activation (CTSC, CASP4, CORO1A), immunometabolism (IDO1, CD36, ELOVL5, MAOA), and epithelial barrier function (LAMB3, ALDH1A3, EGFL6, INTS10). Table 5.1 lists all 15 genes with their functional categories and biological annotations.

Table 5.1: The 15 shared discovery genes with functional category and biological role

Gene Symbol	Functional Category	Biological Relevance
CCL5	Cytokine / Chemokine	T-cell, eosinophil, and monocyte recruitment; binds CCR1/3/5
CCL20	Cytokine / Chemokine	Dendritic cell and lymphocyte recruitment to mucosal surfaces; CCR6 ligand
IL7R	Cytokine Receptor	T-cell development, survival, and homeostasis; IL-7 receptor alpha chain
IDO1	Immune Metabolism	Rate-limiting enzyme of tryptophan catabolism via kynurenine pathway; immune tolerance regulation
CD36	Innate Immune / Lipid	Scavenger receptor; lipid metabolism, fatty acid uptake, and innate immune activation
CASP4	Innate Immune Signalling	Non-canonical inflammasome activation; pyroptotic cell death
CTSC	Protease / Innate Immunity	Cathepsin C; activates neutrophil serine proteases and granzymes
CORO1A	Cytoskeletal / Immune	Actin-binding protein; leukocyte migration and immune synapse formation
PAG1	Immune Signalling Adaptor	Membrane adaptor; fine-tunes T-cell receptor signalling thresholds
ALDH1A3	Metabolic / Epithelial	Aldehyde dehydrogenase; retinoid metabolism and epithelial differentiation
EGFL6	Extracellular Matrix	EGF-like domain protein; tissue remodelling and extracellular matrix organisation
ELOVL5	Lipid Metabolism	Elongation of very long chain fatty acids; lipid mediator and eicosanoid precursor synthesis
INTS10	RNA Processing	Integrator complex subunit; transcriptional regulation of small nuclear RNA

LAMB3	Epithelial Barrier	Laminin subunit beta-3; basement membrane integrity and epithelial adhesion
MAOA	Oxidative Metabolism	Monoamine oxidase A; oxidative catabolism of monoamines; oxidative stress responses

At the strict high-confidence threshold (adjusted $p < 0.05$, $|\log_2FC| \geq 1.0$), four genes — CCL5, CTSC, ELOVL5, and MAOA — were shared across all three datasets. CCL5 was the only gene identified at both the discovery and strict thresholds, confirming it as the most statistically robust shared candidate across all three diseases.

5.3 FUNCTIONAL ENRICHMENT ANALYSIS

The 15 discovery genes were submitted to Metascape for Gene Ontology (GO) biological process and pathway enrichment analysis, integrating GO terms, KEGG pathways, Reactome pathways, and WikiPathways (Zhou *et al.*, 2019). Figure 5.2 displays the Metascape GO biological process enrichment bar plot for the 15 shared discovery genes.

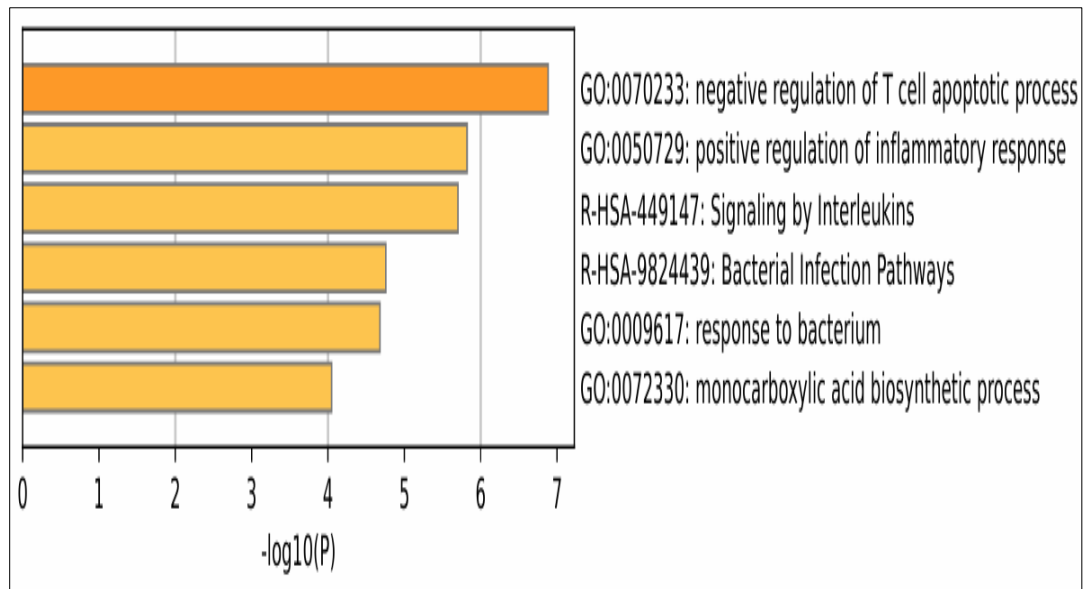


Fig.5.2: Metascape GO Biological Process enrichment bar plot for the 15 shared discovery genes, ranked by significance

The top enriched biological processes included immune system process, cytokine-mediated signalling, T-cell activation, leukocyte chemotaxis, and neutrophil degranulation. Parallel enrichment analysis using g:Profiler confirmed the same dominant biological processes consistently across GO, KEGG, and Reactome databases, applying the g:SCS multiple testing correction (Raudvere *et al.*, 2019). Enrichr further validated these themes across GO Biological Process, Reactome, and KEGG libraries, with cytokine–cytokine receptor interaction (hsa04060) and TNF signalling pathway (hsa04668) among the most significantly enriched KEGG terms (Kuleshov *et al.*, 2016).

To derive a biologically robust gene set supported by multiple enrichment platforms, significance scores from Metascape ($-\log_{10}$ adjusted p-value; log B) and g:Profiler ($-\log_{10}$ g:SCS-corrected p-value) were integrated into a composite scoring table. Genes consistently highly enriched across both platforms were designated as biologically consensus genes. This integration step yielded nine consensus genes: CCL5, CCL20, IL7R, IDO1, CD36, CASP4, CTSC, CORO1A, and PAG1.

5.4 PROTEIN–PROTEIN INTERACTION NETWORK ANALYSIS AND HUB GENE IDENTIFICATION

The nine consensus genes were imported into the STRING database (version 11.5) to construct a protein–protein interaction (PPI) network (Szklarczyk *et al.*, 2019). A medium confidence threshold (combined score ≥ 0.4) was applied, following standard practice for inflammatory disease gene network studies. The resulting network was imported into Cytoscape (v3.10; Otasek *et al.* 2019) for visualisation. Figure 5.3 shows the Cytoscape PPI network of the nine consensus genes, with node size proportional to degree centrality and edge thickness reflecting interaction confidence score.

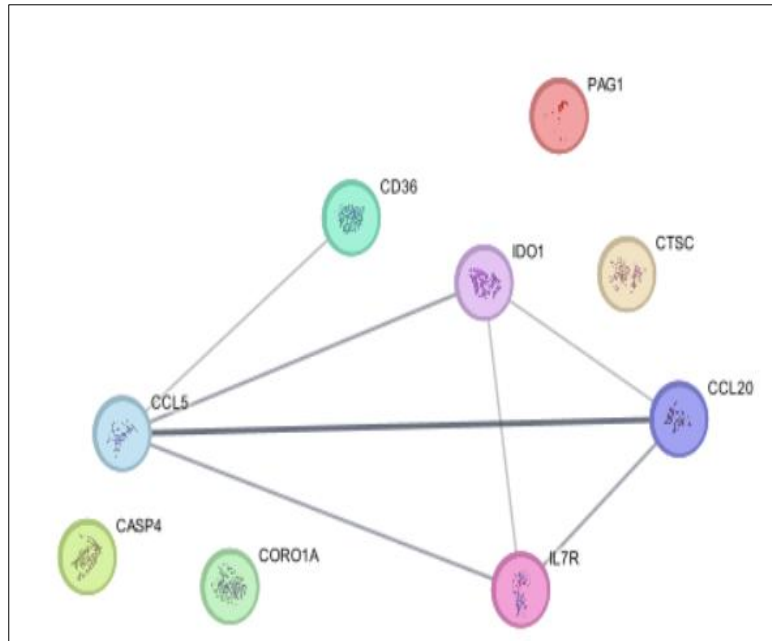


Fig: 5.3: Cytoscape protein–protein interaction network of the 9 consensus genes with node size reflecting degree centrality

Hub gene identification was performed using the CytoHubba plugin applying four centrality measures simultaneously — Maximal Clique Centrality (MCC), Degree, Closeness, and Betweenness (Chin *et al.*, 2014; Tang *et al.* 2015). Figure 5.4 shows the CytoHubba MCC topological analysis, with nodes colour-coded from red (highest rank) to yellow-green (lowest rank).

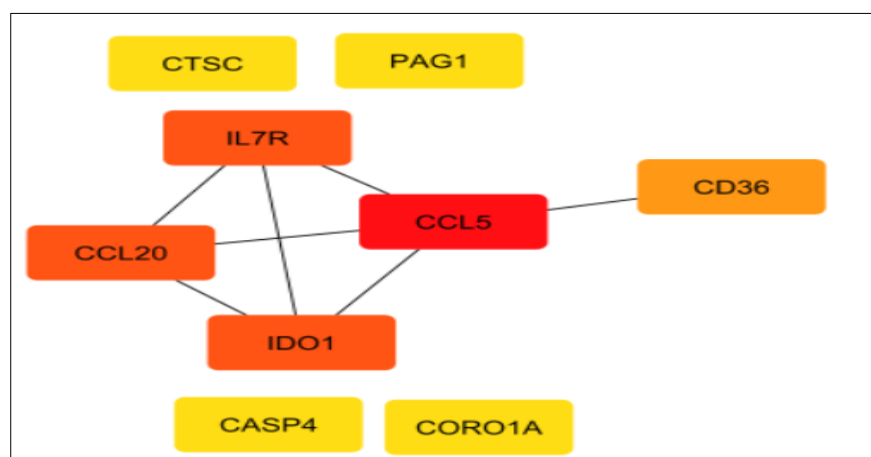


Fig:5.4: CytoHubba MCC topological analysis of the 9 consensus genes — red nodes indicate highest MCC ranking

Genes ranking consistently in the top positions across all four centrality measures were designated as hub genes. Table 5.2 presents the integrated centrality ranking summary for all nine consensus genes. Four genes — CCL5, CCL20, IL7R, and IDO1 — satisfied the multi-centrality criterion and were designated as the hub genes of the shared allergic respiratory disease PPI network.

Table 5.2: CytoHubba centrality ranking for all 9 consensus genes — MCC, Degree, Closeness, and Betweenness ranks

Gene Symbol	MCC (Score / Rank)	Degree (Score / Rank)	Closeness (Score / Rank)	Betweenness (Score / Rank)	Hub Status
↓ Top to Bottom	Dense clique membership	Direct connections	Network reachability	Signalling bottleneck	Top 4 across ALL metrics?
CCL5	7 #1	4 #1	4 #1	6 #1	✓ HUB
CCL20	6 #2	3 #2	3.5 #2	0 #2	✓ HUB
IL7R	6 #2	3 #2	3.5 #2	0 #2	✓ HUB
IDO1	6 #2	3 #2	3.5 #2	0 #2	✓ HUB
CD36	1 #5	1 #5	2.5 #5	0 #2	Non-hub

CASP4	0 #6	0 #6	0 #6	0 #2	Non-hub
CTSC	0 #6	0 #6	0 #6	0 #2	Non-hub
CORO1A	0 #6	0 #6	0 #6	0 #2	Non-hub
PAG1	0 #6	0 #6	0 #6	0 #2	Non-hub

MCC = Maximal Clique Centrality (gold-standard CytoHubba measure for essential hub identification). Degree = number of direct interaction partners. Closeness = ability to reach all network nodes rapidly. Betweenness = regulatory bottleneck control over inter-node communication. Hub designation requires consistent top-4 ranking across ALL four measures.

5.5 Sinusitis Multi-Omics Analysis

Sinusitis (CRSwNP) was selected as the representative disease for multi-omics validation based on the availability of published complementary proteomic and metabolomic datasets for this condition. Proteomic data from a published LC-MS/MS study of CRSwNP nasal lavage fluid yielded 53 significantly upregulated proteins from 230 identified. Metabolomic data from a published untargeted metabolomic profiling study provided 11 disease-associated metabolites for pathway analysis.

Transcriptomic data from GSE23552 were re-analysed for the multi-omics context, and upregulated differentially expressed genes from CRSwNP samples were submitted to Metascape pathway enrichment. Figure 5.5 displays the sinusitis transcriptomic enrichment output.

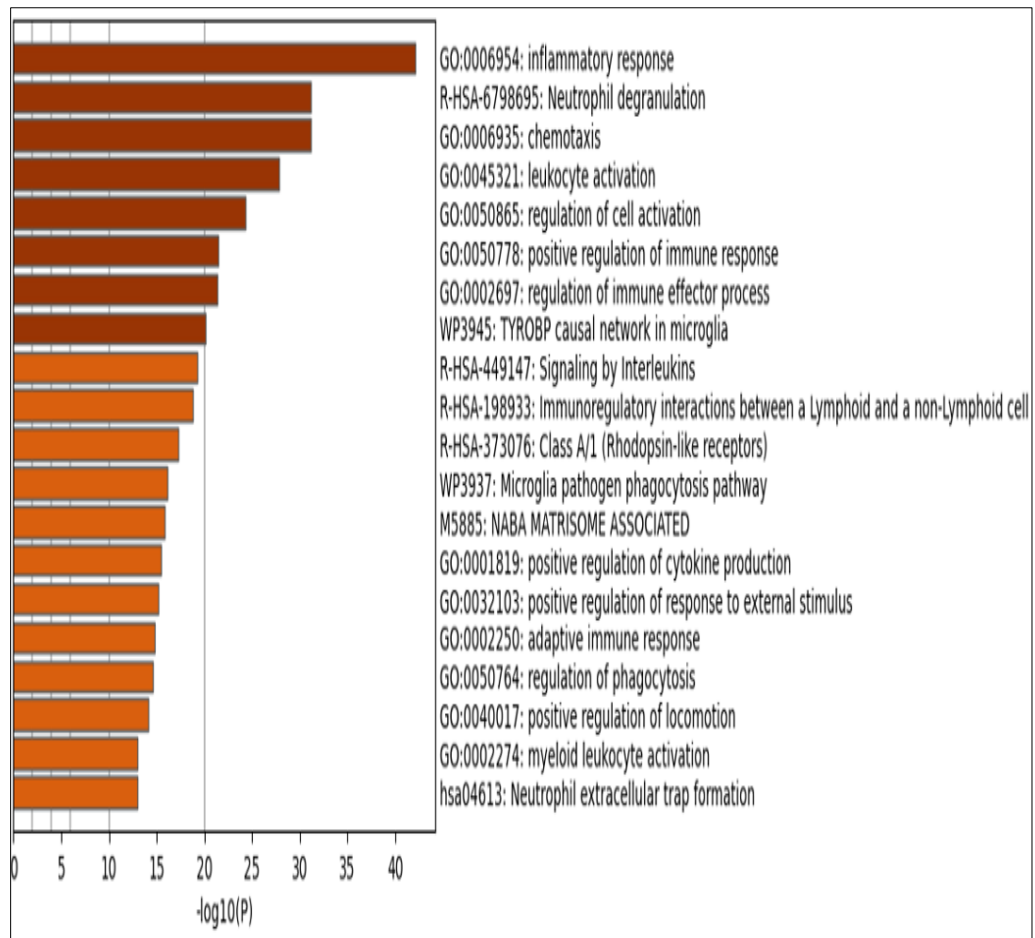


Fig: 5.5: Metascape pathway enrichment for sinusitis upregulated genes from GSE23552

The 53 significantly upregulated proteins from CRSwNP nasal lavage fluid proteomics were submitted to Metascape for pathway enrichment analysis. Figure 5.6 displays the proteomic enrichment output.

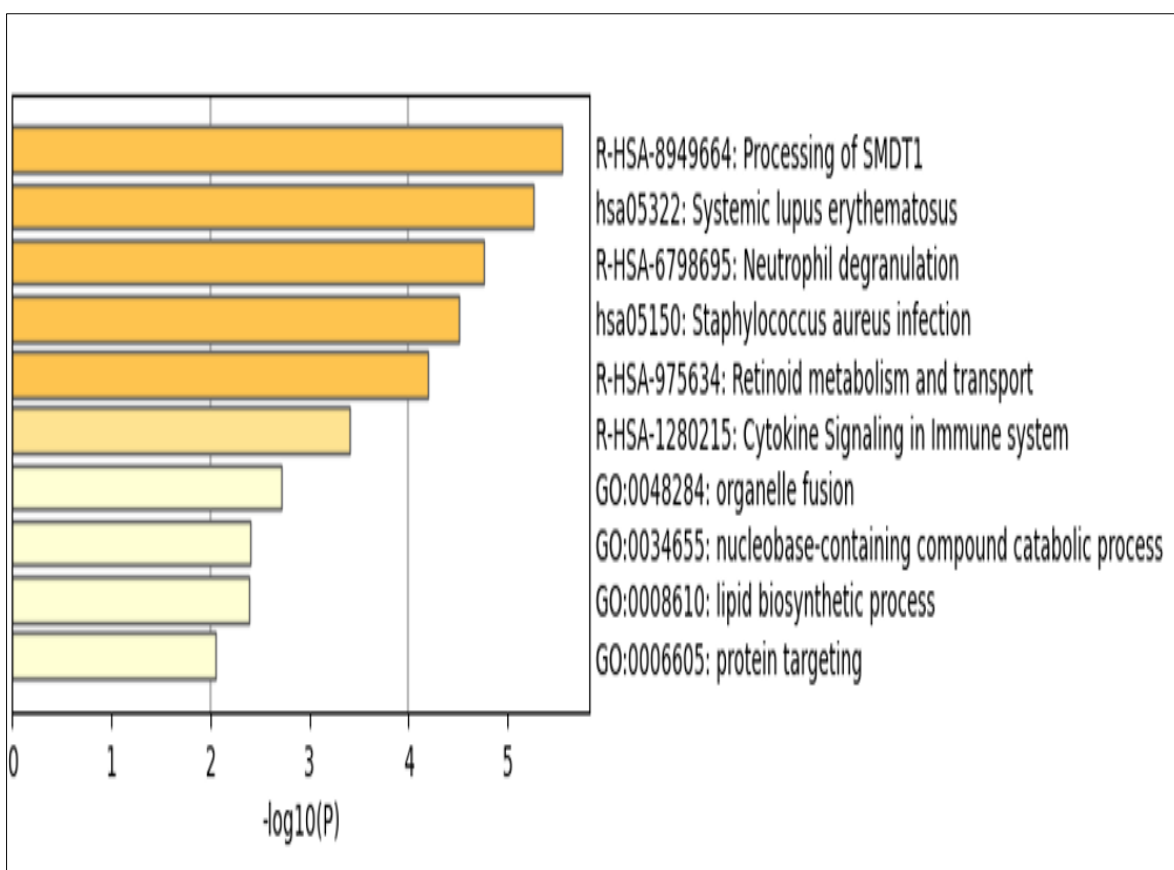


Fig:5.6: Metascape pathway enrichment for 53 upregulated proteins from CRSwNP nasal lavage fluid proteomics

The 11 disease-associated metabolites from the published CRSwNP metabolomics study were submitted to MetaboAnalyst 5.0 (Pang *et al.*, 2021) for pathway enrichment and topology analysis. Figure 5.7 displays the MetaboAnalyst pathway enrichment output.

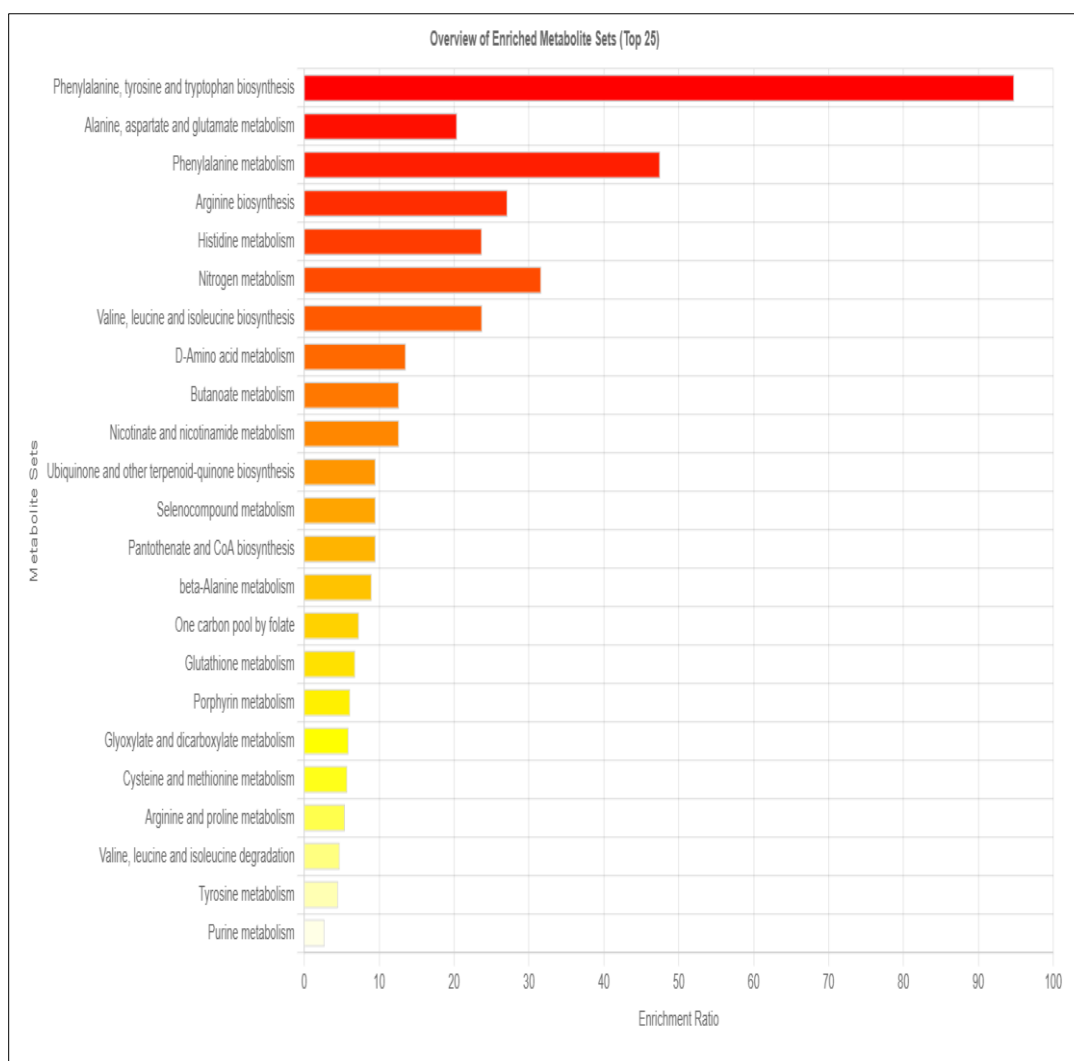


Fig:5.7: MetaboAnalyst pathway enrichment for 11 CRSwNP metabolites — tryptophan and amino acid pathways most enriched

5.6 Quantitative Cross-Omics Pathway Integration

A Python-based computational framework was developed to integrate enriched pathways from all three omics layers into a single quantitative composite convergence score. For each candidate pathway, the composite score was calculated as:

$$\text{Composite Score} = -\log_{10}(\text{adjusted p-value}) \times \text{Combined Score} \times \text{Consistency Score} \times \text{Overlap Percentage}$$

Pathways supported in two or more omics layers were classified as convergent. Figure 5.8 shows the composite convergence scores across all three omics layers.

```

...
OTHER PATHWAYS:

                                pathway
4      Class A/1 (Rhodopsin-like receptors)
24     Processing of SMDT1
35  Valine, leucine and isoleucine degradation

FINAL RESULT:

      Category      RNA  Protein  Metabolite  Combined_Score  \
3      Immune activation  195.91    2.28      0.00      198.19
1      Cytokine signaling   69.71    0.20      0.00      69.91
4      Inflammation        53.08    0.00      0.00      53.08
0      Cellular processes   24.02    0.00      0.00      24.02
6      Other               14.44    1.21      0.13      15.78
5      Metabolic adaptation    0.00    0.93     11.54      12.47
2  Host-pathogen interaction    0.00    2.26      0.00       2.26

Consistency
3          2
1          2
4          1
0          1
6          1
5          1
2          1

OVERLAP %: 28.57142857142857

```

Fig: 5.8: Python quantitative cross-omics pathway integration showing composite convergence scores across all three omics layers

Figure 5.9 presents the cross-omics pathway enrichment heatmap, displaying enrichment significance across transcriptomic, proteomic, and metabolomic layers simultaneously.

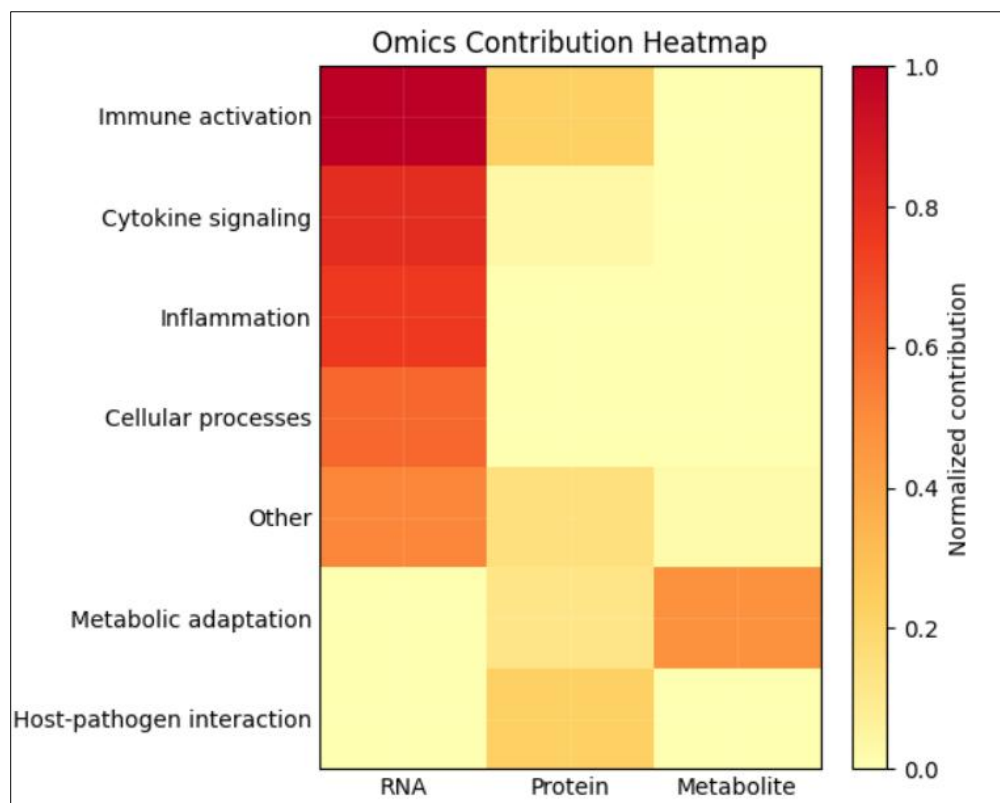


Fig: 5.9: Cross-omics pathway enrichment heatmap showing enrichment significance across transcriptomic, proteomic, and metabolomic layers

Following the composite score computation, a pan-grouped analysis was performed to aggregate individual enriched pathways into broader biological theme categories, enabling a higher-level summary of convergent functional processes across all three omics layers. Pathways were assigned to five theme groups: immune activation and cytokine signalling, innate immunity, adaptive immunity, immune-related metabolism, and amino acid and nitrogen metabolism. Figure 5.10 displays the pan-grouped pathway output. The pathway agreement analysis (Figure 5.11) further confirmed that neutrophil degranulation and cytokine signalling were independently detected in two or more omics layers, representing the most cross-validated convergent themes. This pan-grouped summary directly informed the bridge step comparison presented below.

PAN GROUPED RESULT:		
	Category	Pan_Score
1	Host-pathogen interaction	3.44
2	Immune activation	2.55
3	Inflammation	2.39
0	Cytokine signaling	2.36
4	Metabolic adaptation	1.17


```
print(pan[['pathway','Category']])
```


	pathway	Category
0	negative regulation of T cell apoptotic process	Immune activation
1	positive regulation of inflammatory response	Inflammation
2	Signaling by Interleukins	Cytokine signaling
3	Bacterial Infection Pathways	Host-pathogen interaction
4	response to bacterium	Host-pathogen interaction
5	monocarboxylic acid biosynthetic process	Metabolic adaptation

Fig. 5.10: Pan-grouped pathway output showing dominant biological theme categories across sinusitis multi-omics data

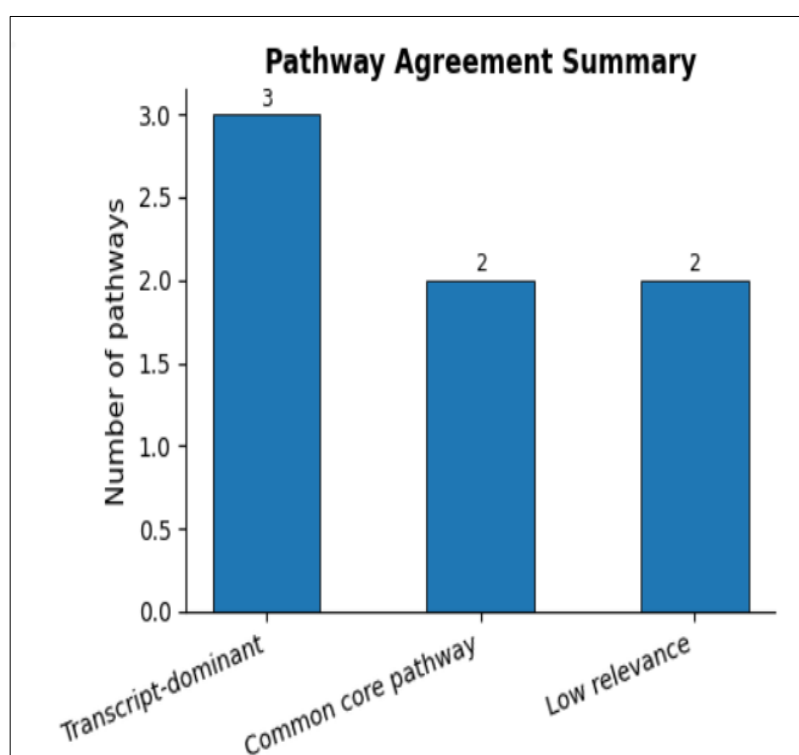


Fig. 5.11: Pathway agreement bar graph showing number of omics layers detecting each biological theme as significantly enriched

The bridge analysis compared pan-transcriptomic pathway signatures (shared across all three diseases) with sinusitis multi-omics integrated pathway findings. Figure 5.12 shows the Reactome Cytokine Signalling in Immune System pathway map with consensus gene positions annotated. Figure 5.13 presents the bar graph comparing biological themes from pan-transcriptomic analysis versus sinusitis multi-omics integration. Concordant themes — including immune activation, cytokine signalling, inflammatory response, leukocyte migration, innate and adaptive immunity, and immune-related metabolism — were confirmed across both analyses.

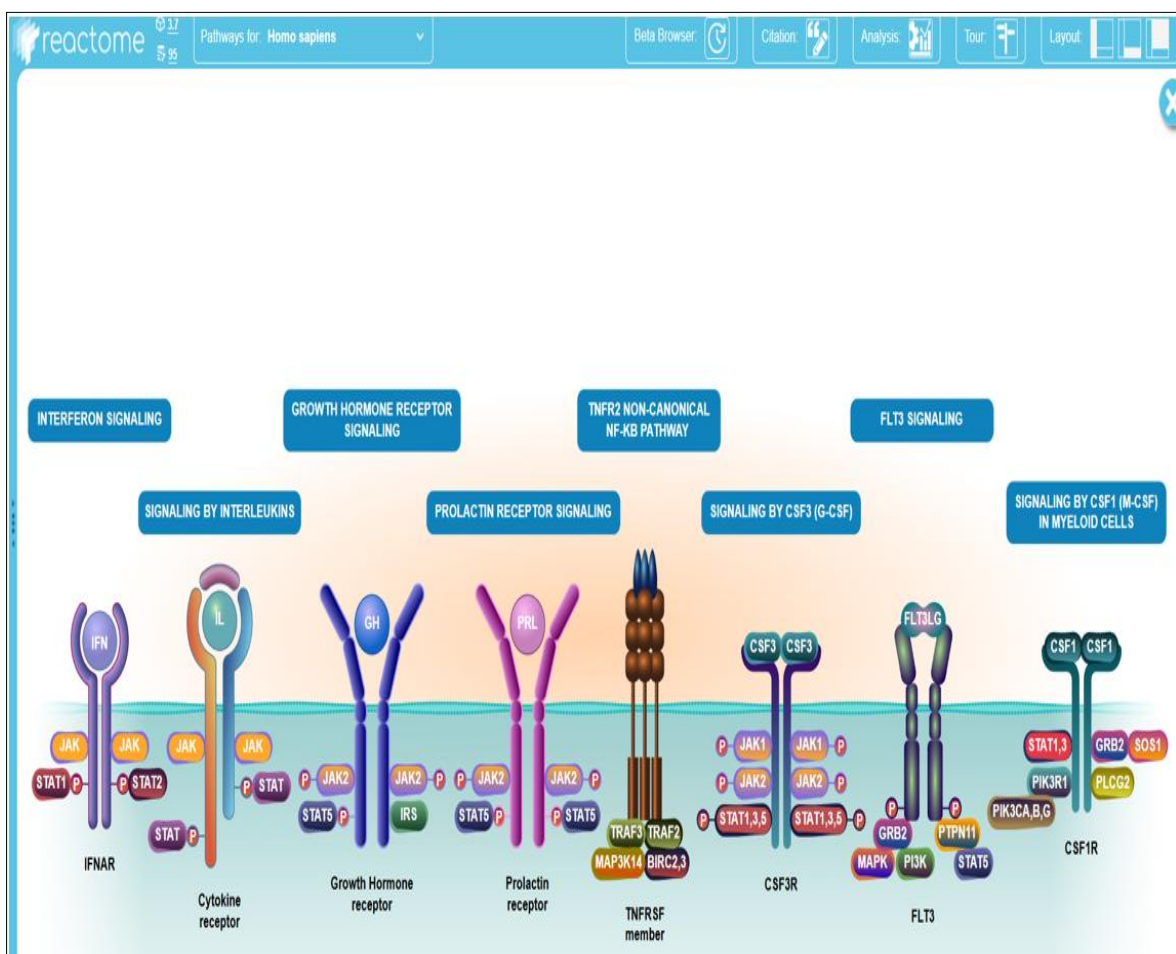


Fig: 5.12: Reactome Cytokine Signalling in Immune System pathway map with consensus gene positions annotated

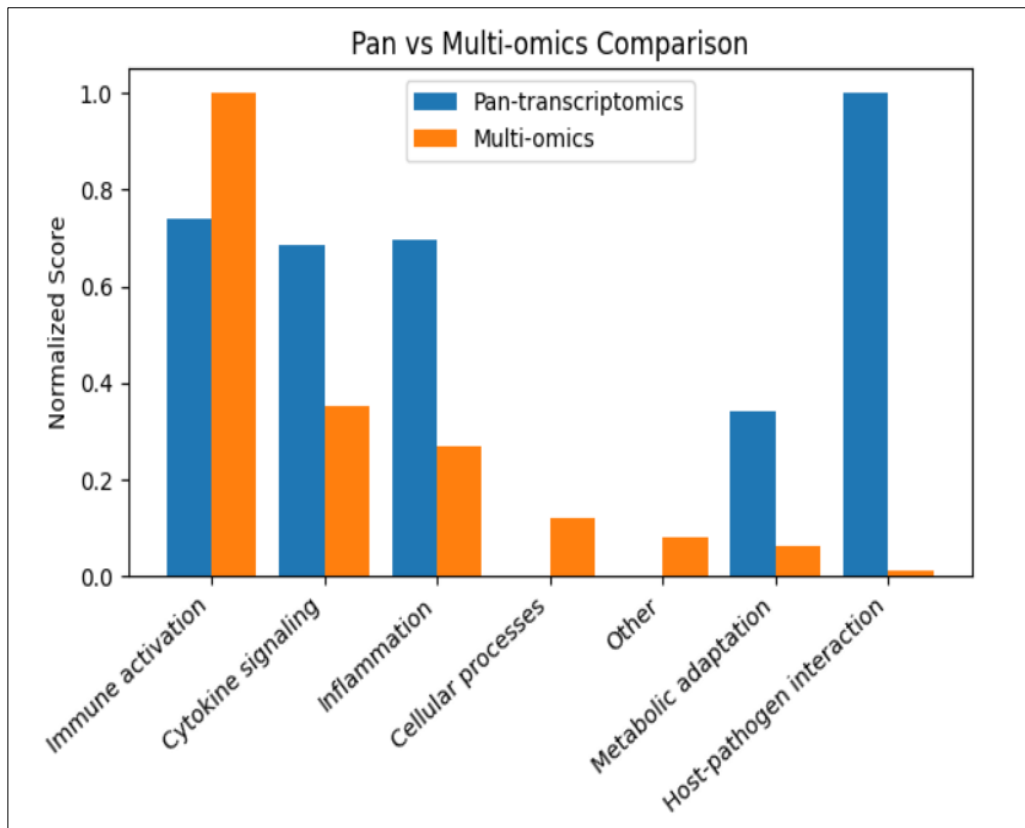


Fig: 5.13: Bar graph comparing biological themes from pan-transcriptomic analysis versus sinusitis multi-omics integration

5.7 Biomarker Finalisation and Validation

The four hub genes — CCL5, CCL20, IL7R, and IDO1 — were finalised as candidate pan-disease biomarkers based on three converging lines of evidence: consistent top-ranking across all four CytoHubba centrality metrics; robust enrichment significance across all three functional annotation platforms; and pathway-level support in at least two of the three omics layers in the sinusitis multi-omics framework. Table 5.3 presents the cross-omics pathway comparison confirming consistency between pan-transcriptomic findings and sinusitis multi-omics results.

Table 5.3: Cross-omics pathway comparison showing pan-transcriptomic findings versus sinusitis multi-omics results

Biological Theme	Pan-Transcriptomic	Transcriptomics (GSE23552)	Proteomics (LC-MS/MS)	Metabolomics (MetaboAnalyst)	Convergent
Immune activation & cytokine signalling	✓	✓	✓	—	Yes (3/3 omics)
Innate immunity / neutrophil degranulation	✓	✓	✓	—	Yes (3/3 omics)
Adaptive immunity / T-cell regulation	✓	✓	—	—	Yes (2/3 omics)
Immune-related metabolism	✓	—	✓	—	Yes (2/3 omics)
Amino acid & tryptophan metabolism	—	—	—	✓	Yes — IDO1 link

Table 5.4 presents the multi-criterion evidence matrix summarising the evidence supporting each of the four candidate biomarkers across all analytical stages.

Table 5.4: Multi-criterion evidence matrix for the four candidate biomarkers

Evidence Criterion	CCL5	CCL20	IL7R	IDO1
Top hub across all 4 CytoHubba metrics	✓	✓	✓	✓
Identified at strict threshold (padj<0.05)	✓	—	—	—
Metascape enrichment (discovery set)	✓	✓	✓	✓
g:Profiler enrichment (discovery set)	✓	✓	✓	✓
Enrichr validation (consensus set)	✓	✓	✓	✓
Transcriptomic omics layer support	✓	✓	✓	✓
Proteomic omics layer support	✓	✓	—	✓
Metabolomic omics layer support (IDO1 link)	—	—	—	✓
Literature validation	✓	✓	✓	✓
External dataset validation (≥1/3)	—	✓	✓	✓

External transcriptomic validation was performed using three independent GEO datasets not used in the primary discovery phase — GSE179265, GSE75011, and GSE72713. Table 5.5 presents the external validation results for all four candidate biomarkers across the three independent datasets.

Table 5.5: External transcriptomic dataset validation results for CCL5, CCL20, IL7R, and IDO1 across three independent datasets

Biomarker	GSE179265	GSE75011	GSE72713	Datasets Validated (of 3)
CCL5	Not significant	Not significant	Not significant	0/3
CCL20	Significant ✓	Significant ✓	Not significant	2/3
IL7R	Significant ✓	Not significant	Not significant	1/3
IDO1	Significant ✓	Not significant	Significant ✓	2/3

5.8 Machine Learning-Based Biomarker Performance Evaluation

Machine learning evaluation was performed using Random Forest (RF) and Logistic Regression (LR) classifiers on the GSE65204 dataset with five-fold stratified cross-validation. A two-part experimental design was applied: Part 1 used all genome-wide gene expression features to establish a discriminative performance baseline; Part 2 used only the four candidate biomarkers (CCL5, CCL20, IL7R, IDO1) as input features to directly evaluate their panel-level classification capacity.

Table 5.6 presents the Part 1 classification performance metrics. Figure 5.14 shows the top 15 gene feature importance from the genome-wide Random Forest analysis. Figure 5.15 shows the ROC curves for both classifiers in Part 1.

Table 5.6: Machine learning Part 1 classification performance metrics — genome-wide analysis on GSE65204

Metric	Random Forest (Part 1)	Logistic Regression (Part 1)
Accuracy	0.857	0.786
AUC (ROC)	0.898	0.939
Precision	0.86	0.79
Recall	0.86	0.79
F1 Score	0.86	0.78
CV Accuracy (5-fold)	0.781	0.752

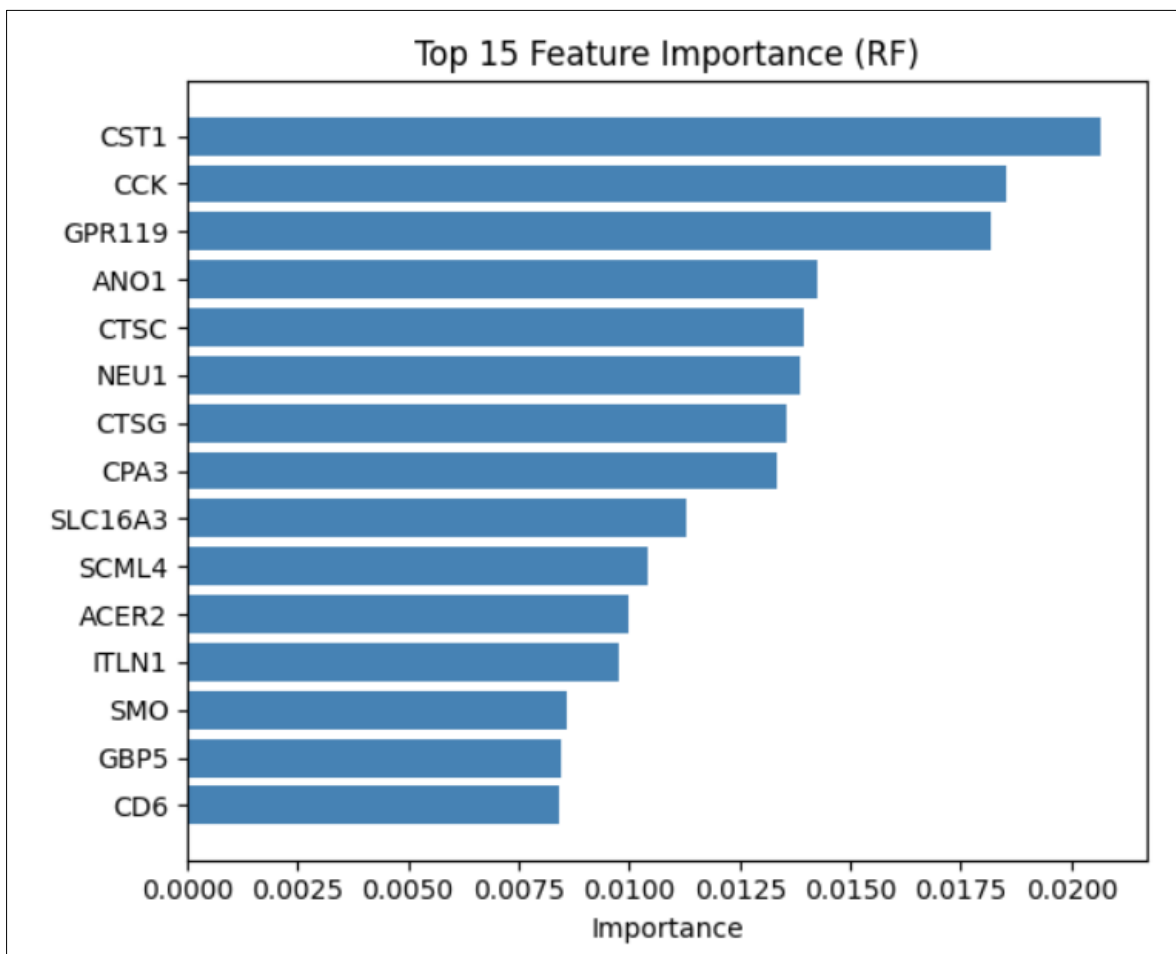


Fig:5.14: Top 15 gene feature importance from genome-wide Random Forest analysis on GSE65204 — Part 1

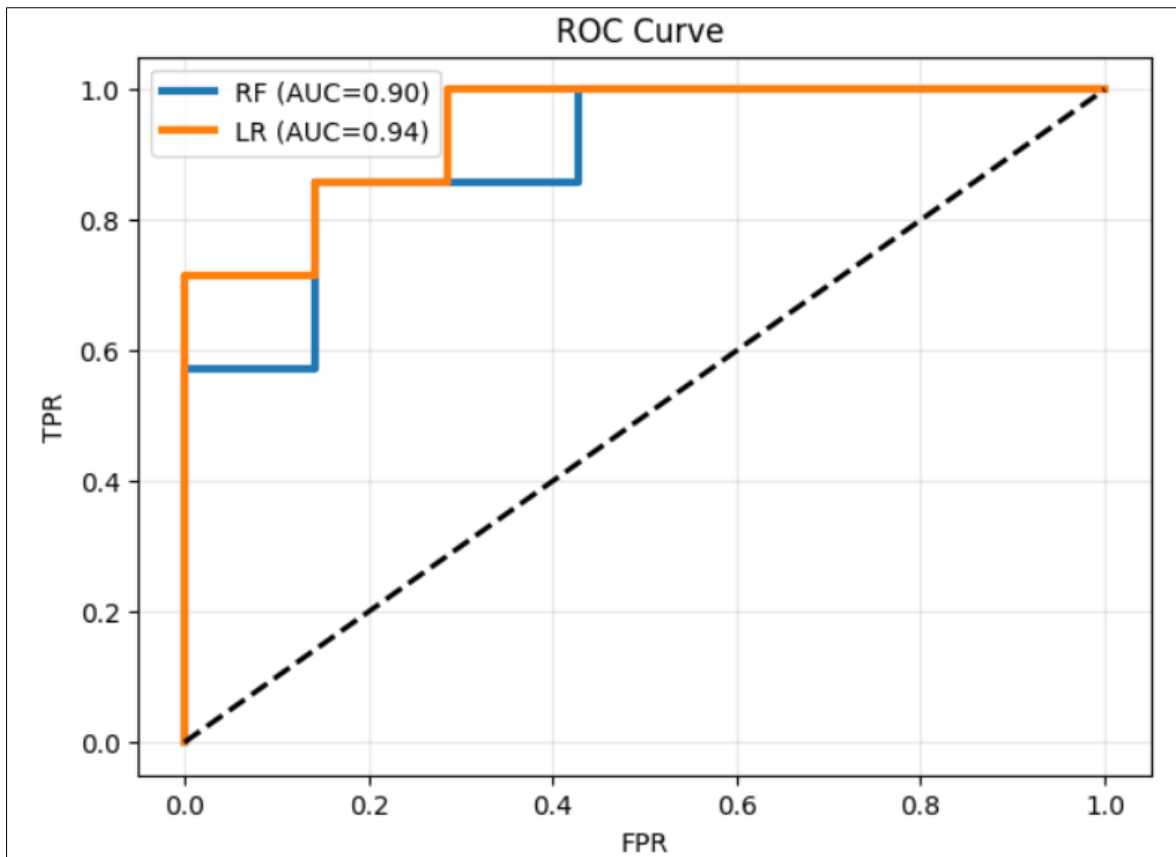


Fig. 5.15: ROC curves for Random Forest and Logistic Regression classifiers — Part 1 genome-wide analysis

Table 5.7 presents the Part 2 classification performance metrics using only the four candidate biomarkers as features. Figures 5.16 and 5.17 show the ROC curves and feature importance bar graph for the four-biomarker panel.

Table 5.7: Machine learning Part 2 classification performance metrics — four-biomarker panel on GSE65204

Metric	Random Forest (Part 2)	Logistic Regression (Part 2)
Accuracy	0.643	0.714
AUC (ROC)	0.776	0.816
Precision	0.65	0.71
Recall	0.64	0.71
F1 Score	0.64	0.71
CV Accuracy (5-fold)	0.781	0.752

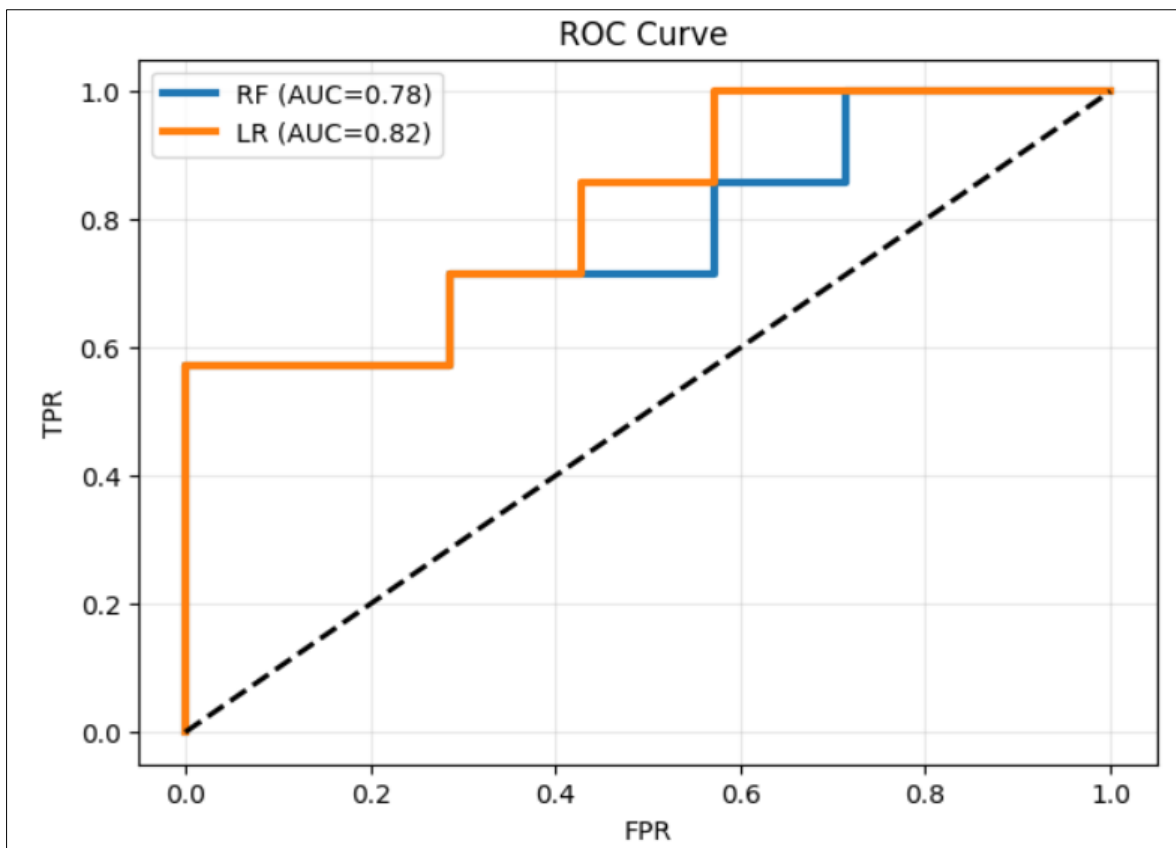


Fig: 5.16: ROC curves for Random Forest and Logistic Regression classifiers — Part 2 four-biomarker panel

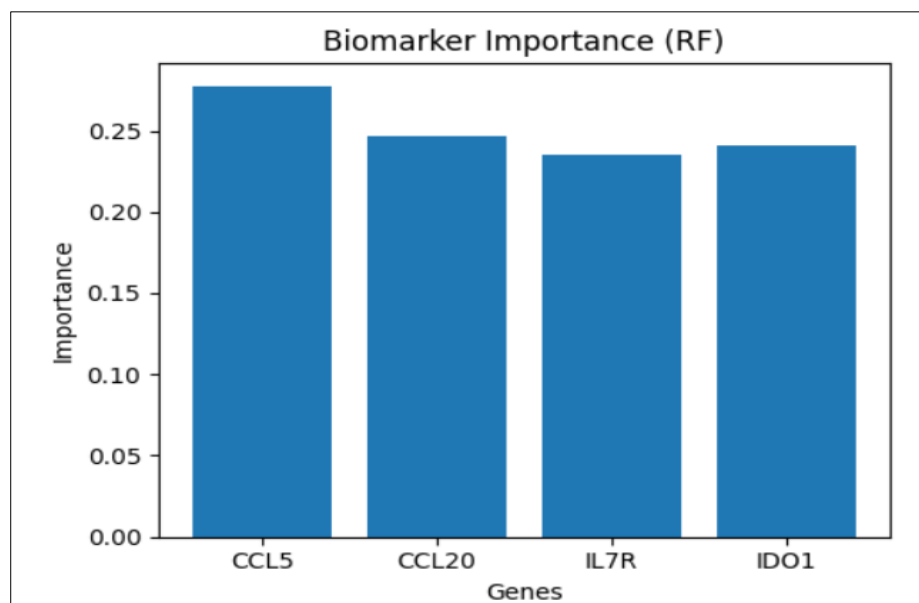


Fig: 5.17: Relative feature importance of the four candidate biomarkers from Random Forest classifier — Part 2

The genome-wide Part 1 analysis achieved AUC values of 0.898 (RF) and 0.939 (LR), confirming strong discriminative performance across the GSE65204 dataset. The constrained four-biomarker Part 2 panel achieved AUC values of 0.776 (RF) and 0.816 (LR) with five-fold cross-validated accuracy of 0.781 and 0.752 respectively, confirming that the candidate panel retains meaningful discriminative capacity above chance levels. None of the four candidate biomarkers appeared in the top 15 genome-wide feature importance genes from Part 1, which is expected and biologically interpretable: the candidates capture cross-disease generalisable signals that differ from single-dataset statistical dominance (Peng *et al.*, 2021).

5.9 DISCUSSION

The decision to use publicly archived GEO microarray datasets for pan-transcriptomic discovery reflects the established scientific value of systematically reanalysing high-quality public data as a rigorous and ethically efficient complement to primary data generation (Rosati *et al.*, 2024). Processing three independent datasets — GSE9150, GSE23552, and GSE65204 — using a uniform RMA normalisation and limma differential expression pipeline, applied independently within each dataset, mitigates batch confounding and ensures that shared gene signatures represent genuine cross-disease signals rather than platform-specific artefacts (Ritchie *et al.*, 2015; Luo *et al.*, 2015).

The 15 shared discovery genes encode functionally diverse proteins spanning immune cell recruitment (CCL5, CCL20), T-cell homeostasis (IL7R, PAG1), innate immune activation (CTSC, CASP4, CORO1A), immunometabolism (IDO1, CD36, ELOVL5, MAOA), and epithelial barrier maintenance (LAMB3, ALDH1A3, EGFL6, INTS10) — a multi-layer biological landscape consistent with the united airways concept (Bousquet *et al.*, 2019). The consistent identification of cytokine-mediated signalling, T-cell activation, and neutrophil degranulation as the top enriched themes across all three independent enrichment platforms (Metascape, g:Profiler, and Enrichr) eliminates the possibility that these findings are tool-specific artefacts (Zhou *et al.*, 2019; Raudvere *et al.*, 2019; Kuleshov *et al.*, 2016). The enrichment of neutrophil degranulation alongside CTSC — a direct activator of neutrophil serine proteases — indicates that neutrophil-mediated innate immunity may be a

more fundamental shared mechanism in allergic disease than the classically emphasised eosinophilic Th2 response (Ha and Kim, 2025).

The application of four CytoHubba centrality measures simultaneously — MCC, Degree, Closeness, and Betweenness — provides multi-metric robustness to hub gene designation, ensuring that CCL5, CCL20, IL7R, and IDO1 are topologically prominent by every measurable criterion rather than by any single algorithmic assumption (Chin et al., 2014). Each hub gene carries direct biological and clinical evidence: CCL5 orchestrates recruitment of T lymphocytes, eosinophils, and NK cells to allergic airway tissue through CCR1/3/5 interactions (Dorismond et al., 2025); CCL20 directs dendritic cells and Th17 cells to mucosal surfaces via CCR6, with experimental evidence confirming its role in amplifying airway inflammatory responses (Li et al., 2024); IL7R blockade significantly reduces Th2 cytokine production and eosinophilic infiltration in experimental allergic models (Le Mai et al., 2020); and IDO1 encodes the rate-limiting enzyme of the tryptophan–kynurenine pathway, positioning immunometabolic reprogramming as a shared regulatory dimension of allergic disease biology (Miao et al., 2024).

Multi-omics integration confirmed that the shared transcriptional immune signatures are functionally active at the protein level (cytokine signalling and neutrophil degranulation in the CRSwNP proteome) and at the metabolite level (tryptophan catabolism and amino acid metabolism in the CRSwNP metabolome), addressing the well-documented 40–60% mRNA-to-protein correlation gap (Chen et al., 2023; Subramanian et al., 2020). The IDO1–tryptophan cross-omics link — a hub gene identified transcriptionally and independently confirmed metabolomically — represents the most mechanistically specific finding of this study. Variable external validation (CCL20 and IDO1 supported in 2/3 datasets; IL7R in 1/3; CCL5 in 0/3) reflects known biological heterogeneity across disease endotypes, cohort compositions, and tissue types rather than biomarker invalidity (Rosati et al., 2024; Dorismond et al., 2025). Machine learning evaluation confirmed that the four-biomarker panel retains meaningful discriminative capacity (AUC 0.776–0.816 in the constrained Part 2 model), while the absence of the candidates from Part 1 genome-wide feature importance rankings demonstrates that multi-omics integration captures biologically generalisable signals that single-dataset statistical ranking alone would not prioritise (Peng et al., 2021; Subramanian et al., 2020).

The principal limitations of the study include restriction of multi-omics validation to sinusitis alone due to data availability, the use of literature-curated proteomic and metabolomic data from different patient cohorts than the transcriptomic datasets, the inherent dynamic range limitations of microarray-based transcriptomics relative to RNA sequencing, partial circularity in the machine learning validation arising from reuse of GSE65204, and the absence of experimental protein-level validation in patient samples — all of which appropriately scope the interpretation within the context of a computational investigation (Ritchie et al., 2015; Agache et al., 2025; Peng et al., 2021).

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 Summary

This study sought to provide an answer to a question that has seldom been addressed systematically in the molecular biology of allergic disease: what molecular pathways are common between dust allergy, CRSwNP, and allergic asthma, and whether such common pathways could be confirmed not only transcriptionally but also at the protein and metabolite levels. The study was conducted entirely in silico, using publicly archived microarray gene expression data from three GEO datasets — GSE9150 (dust allergy), GSE23552 (CRSwNP), and GSE65204 (allergic asthma) — supplemented by published proteomic and metabolomic datasets from CRSwNP for the multi-omics validation phase.

The analytical pipeline was structured across eight sequential stages: transcriptomic data processing and differential expression analysis using a standardised R pipeline; cross-disease gene overlap at two significance thresholds; functional enrichment analysis across three independent platforms (Metascape, g:Profiler, Enrichr); PPI network construction and hub gene identification in STRING and Cytoscape using four CytoHubba centrality measures; sinusitis-specific multi-omics integration across transcriptomic, proteomic, and metabolomic layers; quantitative cross-omics pathway integration using a Python-based composite scoring framework; two-step biomarker validation through literature review and three independent external datasets; and machine learning evaluation using Random Forest and Logistic Regression classifiers with five-fold cross-validation.

6.2 Principal Findings

- **Shared transcriptional signature-** Gene expression overlap analysis identified 15 genes differentially expressed across all three allergic respiratory diseases at the discovery threshold, spanning chemokine-mediated immune recruitment (CCL5, CCL20), T-cell regulation (IL7R,

PAG1), innate immune activation (CTSC, CASP4, CORO1A), immunometabolism (IDO1, CD36, ELOVL5, MAOA), and epithelial barrier function (LAMB3, ALDH1A3, EGFL6, INTS10). Four genes — CCL5, CTSC, ELOVL5, and MAOA — were confirmed at the strict high-confidence threshold, with CCL5 identified at both levels.

- **Convergent biological pathways-** Functional enrichment analysis using Metascape, g:Profiler, and Enrichr consistently identified immune system process, inflammatory response, cytokine-mediated signalling, T-cell activation, and neutrophil degranulation as the most robustly enriched shared biological processes — with consistent findings across all three platforms confirming the biological robustness of these themes (Reimand *et al.*, 2019).
- **Hub gene identification-** Protein–protein interaction network analysis confirmed that the 9 consensus genes form an interconnected molecular network. CytoHubba topological scoring across MCC, Degree, Closeness, and Betweenness uniformly identified CCL5, CCL20, IL7R, and IDO1 as the four hub genes — consistently placed in the top positions across all four metrics, substantially reducing the probability of false-positive hub designation (Chin *et al.*, 2014; Tang *et al.* 2015; Gysi *et al.* 2021).
- **Multi-omics corroboration-** Multi-omics analysis of sinusitis confirmed that the shared transcriptional immune themes are functionally active at the protein level (cytokine signalling, neutrophil degranulation) and the metabolite level (tryptophan catabolism, amino acid metabolism). The IDO1–tryptophan cross-omics link — a hub gene identified transcriptionally, confirmed metabolically — represents the most mechanistically specific finding of the study.
- **Biomarker validation and machine learning-** CCL20 and IDO1 were validated in two of three external datasets; IL7R in one. In the machine learning evaluation, Part 1 (genome-wide) achieved AUC values of 0.898 (Random Forest) and 0.939 (Logistic Regression). Part 2 (biomarker-

specific: CCL5, CCL20, IL7R, IDO1 only) achieved AUC values of 0.776 (Random Forest) and 0.816 (Logistic Regression) with five-fold cross-validated accuracy of 0.781 and 0.752 respectively — confirming that the four-candidate panel retains meaningful discriminative capacity as a constrained biological model (Breiman, 2001).

6.3 Conclusion

This study has demonstrated, through a rigorous and systematically structured integrative analysis, that immune activation, cytokine signalling, neutrophil-mediated innate immunity, and chemokine-driven leukocyte recruitment are the defining shared biological themes of allergic respiratory disease at the molecular level — and that these themes are not merely transcriptional signals but functionally supported processes detectable at the protein and metabolite levels in sinusitis tissue. The four candidate biomarkers — CCL5, CCL20, IL7R, and IDO1 — represent the most biologically credible pan-disease targets identified by this pipeline, each supported by convergent evidence from network topology, pathway enrichment, multi-omics layer analysis, published literature, external transcriptomic datasets, and machine learning classification.

The integrative framework developed and applied here — combining pan-transcriptomic discovery, multi-tool functional enrichment, network topology, multi-omics validation, and machine learning evaluation — provides a more reliable and more biologically meaningful basis for candidate prioritisation than any isolated approach could offer. While further experimental and clinical validation remains necessary before these candidates can be considered for diagnostic or therapeutic application, the biological case for CCL5, CCL20, IL7R, and IDO1 as shared molecular mediators of allergic respiratory inflammation is now more comprehensively documented than it has been before — and establishes a foundation on which future experimental, translational, and clinical studies can build.

CHAPTER 7

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ANNEXURES 1

PLAGIARISM REPORT

Page 2 of 71 - Integrity Overview

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ANNEXURES 2

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ANNEXURES 3

SDG MAPPING

SDG	JUSTIFICATION
SDG 3	The study directly addresses Good Health and Well-being by investigating shared molecular mechanisms underlying dust allergy, chronic rhinosinusitis with nasal polyps (CRSwNP), and allergic asthma — three chronic non-communicable respiratory diseases that collectively affect millions globally, with rising prevalence driven by urbanisation and air pollution.
SDG 3	Pan-transcriptomic analysis of three NCBI GEO microarray datasets (GSE9150, GSE23552, GSE65204) identified 15 differentially expressed genes shared across all three allergic respiratory diseases, providing a molecular basis for understanding their co-occurrence and shared immunopathology in clinical populations.
SDG 3	Protein–protein interaction (PPI) network analysis using STRING and Cytoscape identified four hub biomarkers — CCL5, CCL20, IL7R, and IDO1 — as key molecular regulators of cytokine-mediated inflammation, immune cell recruitment, and T-cell homeostasis in allergic airway disease, contributing to therapeutic target identification.
SDG 3	Functional enrichment analysis using Metascape, g:Profiler, and Enrichr identified convergent biological processes including cytokine-mediated signalling, T-cell activation, leukocyte chemotaxis, and neutrophil degranulation as shared pathological themes across dust allergy, CRSwNP, and allergic asthma, advancing disease mechanism understanding.
SDG 3	Multi-omics integration of transcriptomic, proteomic (LC-MS/MS nasal lavage fluid), and metabolomic data for CRSwNP validated that the identified shared immune pathways are functionally active at the protein and metabolite levels, strengthening the translational

	relevance of the biomarker candidates for clinical application.
SDG 3	IDO1 — identified as a hub gene via CytoHubba MCC centrality — was cross-validated by tryptophan–kynurenine metabolic pathway enrichment in the CRSwNP metabolome, establishing an immunometabolic mechanism linking chronic airway inflammation to immune tolerance dysregulation, directly relevant to therapeutic development.
SDG 3	Machine learning classifiers (Random Forest and Logistic Regression) achieved AUC values of 0.898 and 0.939 respectively in genome-wide analysis, and retained meaningful discriminative performance using only the four candidate biomarkers, demonstrating their potential for computationally assisted disease classification and precision diagnostics.
SDG 3	External validation of hub biomarkers across three independent GEO datasets (GSE179265, GSE75011, GSE72713) confirmed that CCL20 and IDO1 show statistically significant differential expression in 2/3 independent cohorts, supporting the reproducibility and generalisability of the identified biomarkers for respiratory disease diagnosis.
SDG 3	The elective course Cancer Biology and Therapeutics (SBIB54004) provided foundational knowledge of disease-driving molecular pathways, tumour microenvironment immunology, and therapeutic target identification strategies — directly informing the approach used in this study for biomarker prioritisation and pathway-level disease characterisation.
SDG 3	The elective course Bioethics, Biosafety and Intellectual Property Rights (SBIB54001) reinforced responsible research practices, including ethical use of publicly available patient-derived transcriptomic datasets, data privacy considerations in computational biomedical research, and responsible reporting of biomarker findings with clinical implications.
SDG 4	The M.Sc. Bioinformatics and Data Science programme integrates

	Quality Education through a multi-semester curriculum spanning biological foundations, computational methods, machine learning, and multi-omics analysis — equipping students with transdisciplinary skills to address complex biomedical research problems.
SDG 4	The R programming pipeline developed for pan-transcriptomic analysis — incorporating GEOquery, limma, and RMA normalisation — demonstrates the direct application of computational skills taught in courses such as R for Bioinformatics (SBIB5301) and Bioinformatics programming (SBIB5102), translating classroom learning into real research outputs.
SDG 4	Python-based quantitative cross-omics pathway integration and machine learning model development using scikit-learn directly applies skills from Python Programming for Bioinformatics (SBIB5102), Next Generation Sequence Data Analysis (SBIB5201), and Machine Learning for Bioinformatics (SBIB5302) to a real-world biomedical research problem.
SDG 4	The study's reproducible eight-stage computational pipeline — from raw data acquisition through biomarker validation and machine learning evaluation — models scientific rigour and open-science practices that are central to quality education in bioinformatics, preparing students for research, industry, and healthcare data roles.
SDG 9	The development of a custom Python-based quantitative cross-omics pathway integration framework — computing composite convergence scores across transcriptomic, proteomic, and metabolomic layers — represents an innovative computational infrastructure contribution to biomedical data science and multi-omics analytical methodology.
SDG 9	Reanalysis of publicly archived microarray datasets (NCBI GEO) through standardised, reproducible pipelines demonstrates how open-access biological data infrastructure can be leveraged to generate new scientific insights at no experimental cost, supporting evidence-based, resource-efficient innovation in biomedical

	research.
SDG 9	Application of machine learning models — Random Forest ensemble classifiers and Logistic Regression — for transcriptomic biomarker evaluation contributes to the advancement of AI-driven diagnostic infrastructure in precision medicine, aligning with industry innovation goals in healthcare data analytics.
SDG 9	The use of established bioinformatics tools (STRING, Cytoscape, CytoHubba, Metascape, g:Profiler, MetaboAnalyst) within an integrated analytical pipeline models scalable, reproducible research infrastructure that can be extended to other inflammatory and non-communicable disease contexts, advancing industrial and clinical research capacity.
SDG 9	The Systems Biology course (SBIB5202) and Next Generation Sequence Data Analysis Lab (SBIB6201) provided training in large-scale biological data handling and network modelling — skills directly applied in constructing the PPI network and pathway convergence framework used in this study.
SDG 17	This study operationalises Partnerships for the Goals by integrating publicly deposited datasets from independent international research groups (GSE9150, GSE23552, GSE65204) with published proteomic and metabolomic datasets from peer-reviewed literature, demonstrating the scientific value of open-access data sharing and global research collaboration.
SDG 17	Cross-validation of biomarker findings across three independent external GEO datasets from distinct patient cohorts, laboratories, and disease contexts exemplifies the principle of multi-institutional data partnership, reinforcing reproducibility and the global generalisability of the identified molecular signatures.

ANNEXURES 4
PO–PSO Mapping
TableA3.1 PO–PSO Mapping

PO/PSO	JUSTIFICATION
PO 1	Advanced and specialised knowledge in bioinformatics and computational biology was critically applied to investigate shared molecular mechanisms across dust allergy, chronic rhinosinusitis with nasal polyps (CRSwNP), and allergic asthma. Multi-omics data spanning transcriptomics, proteomics, and metabolomics were integrated to analyse complex immunological systems underlying allergic respiratory diseases.
PO 2	A reproducible eight-stage computational research pipeline was designed from scratch, integrating R-based transcriptomic preprocessing (GEOquery, limma, RMA normalisation), multi-platform functional enrichment (Metascape, g:Profiler, Enrichr), PPI network construction (STRING, Cytoscape, CytoHubba), and a custom Python-based quantitative cross-omics pathway integration framework with composite convergence scoring, to address complex cross-disease biological questions.
PO 3	Proficiency was demonstrated in selecting and deploying an advanced suite of bioinformatics tools: R/Bioconductor (limma, GEOquery), Python (scikit-learn, pandas), STRING v11.5, Cytoscape v3.10, CytoHubba, Metascape, g:Profiler, Enrichr, MetaboAnalyst 5.0, and Reactome. Machine learning frameworks including Random Forest and Logistic Regression were implemented for biomarker discriminative evaluation with five-fold cross-validation.
PSO 2	A protein–protein interaction (PPI) network was constructed using the STRING database (v11.5) at medium confidence (score ≥ 0.4) and analysed in Cytoscape (v3.10) using four simultaneous topological centrality measures — MCC, Degree, Closeness, and

	Betweenness — via the CytoHubba plugin. This network medicine approach identified CCL5, CCL20, IL7R, and IDO1 as hub regulators, demonstrating the ability to understand disease mechanisms at a systems biology level.
PSO 3	A complete end-to-end bioinformatics workflow was designed and implemented: (1) transcriptomic data acquisition from NCBI GEO; (2) preprocessing with RMA normalisation and PCA QC; (3) differential expression analysis using limma; (4) cross-disease gene overlap; (5) multi-platform functional enrichment; (6) PPI network construction and hub gene identification; (7) multi-omics pathway integration using a custom Python composite convergence scoring framework; and (8) machine learning-based biomarker evaluation — representing a complete translational bioinformatics pipeline for real-world biomedical applications.

PEO–PO Mapping
Table A3.2 PEO–PO Mapping

PO/PEO	JUSTIFICATION
PEO 1	Advanced computational and analytical competency was attained by developing a sophisticated multi-omics integration pipeline combining transcriptomics, proteomics, and metabolomics data for allergic respiratory disease research. Application of R/Bioconductor, Python, and machine learning frameworks to large-scale GEO datasets demonstrates preparedness for research and biotechnology industry roles requiring expert bioinformatics capabilities.
PEO 2	The study establishes interdisciplinary research competence by integrating immunology, genomics, network biology, and data science within a single cohesive project. Identification of four cross-disease hub biomarkers (CCL5, CCL20, IL7R, IDO1) with clinical and therapeutic relevance, communicated through structured scientific writing and rich visualisation, demonstrates the analytical and communication skills required for research, academic, and industry careers.
PEO 3	The project reflects continuous professional growth by adopting state-of-the-art emerging technologies including AI-based machine learning classifiers (Random Forest, Logistic Regression), multi-omics integration frameworks, and cross-omics composite convergence scoring — paired with responsible research conduct through ethical use of open-access data, reproducible workflows, and transparent reporting of biomarker findings.
PO 1	Advanced domain knowledge in computational biology and bioinformatics was applied to analyse three complex allergic respiratory disease datasets simultaneously, integrating immunological knowledge of type 2 immune responses, cytokine signalling, eosinophilic inflammation, and the united airways concept to contextualise molecular findings within a clinically meaningful framework.

PO 2	Problem-solving and research design skills were demonstrated through the construction of a novel quantitative cross-omics pathway integration framework, the development of consensus gene derivation methodology integrating outputs from multiple independent enrichment platforms, and the design of a two-part machine learning evaluation scheme to separately assess genome-wide and biomarker-panel discriminative performance.
PO 3	Proficiency in modern bioinformatics tools and technologies was achieved through hands-on implementation of a comprehensive pipeline spanning R/Bioconductor (limma, GEOquery), Python (scikit-learn, pandas, matplotlib), STRING, Cytoscape, CytoHubba, Metascape, g:Profiler, Enrichr, MetaboAnalyst 5.0, and Reactome — all customised and orchestrated within a single, reproducible analytical workflow.
PO 4	Effective communication of advanced scientific findings was achieved through structured presentation of results using diverse visualisation techniques: Venn diagrams, GO enrichment bar plots, PPI network diagrams, CytoHubba colour-coded topological maps, cross-omics pathway heatmaps, ROC curves, and feature importance bar charts — designed to communicate multi-layered computational findings to diverse scientific audiences.
PO 5	Independent research and problem-solving skills were demonstrated through the complete self-directed execution of an eight-stage analytical pipeline, navigating methodological decisions such as threshold selection, multi-platform consensus derivation, and external dataset validation design, while following a structured and reproducible scientific workflow.
PO 6	Ethical standards were upheld by using only publicly available, properly cited open-access datasets, maintaining full reproducibility through a documented standardised pipeline, avoiding access to patient-identifiable information, and responsibly interpreting and reporting candidate biomarker findings — consistent with professional ethical obligations in bioinformatics and biomedical data science research.

APPENDIX

WORKING CODE

This appendix presents representative code screenshots for each major analytical stage. Full code for all three diseases and complete scripts are available from the author upon request.

```
1 #
2 # GSE9150 - Dust Mite Allergy vs Control
3 #
4 # 0. Install & load packages
5 if (!requireNamespace("GEOquery")) install.packages("GEOquery")
6 if (!requireNamespace("limma")) install.packages("limma")
7 if (!requireNamespace("ggplot2")) install.packages("ggplot2")
8 library(GEOquery)
9 library(limma)
10 library(ggplot2)
11 # 1. Download dataset
12 gset <- getGEO("GSE9150", GSEMatrix = TRUE, getGPL = FALSE)[[1]]
13 expr <- exprs(gset)
14 pheno <- pData(gset)
15 # 2. Define groups correctly
16 pheno$Group <- ifelse(
17   pheno$characteristics_ch1 == "non allergic",
18   "Control",
19   "DustMite"
20 )
21 pheno$Group <- factor(pheno$Group, levels = c("Control", "DustMite"))
22 cat("\nGroup counts:\n")
23 print(table(pheno$Group, useNA = "ifany"))
24 # Save phenotype info
25 write.csv(pheno, "GSE9150_pheno.csv", row.names = TRUE)
26 # 3. Log2 transform if necessary
27 qx <- as.numeric(quantile(expr, c(0, .25, .5, .75, 1.0), na.rm = TRUE))
28 logc <- (qx[5] > 100) || (qx[2] > 0 && qx[2] < 1 && qx[4] > 1 && qx[4] < 2)
29 if (logc) {
30   expr <- log2(expr + 1)
31 }
32 # Save normalized expression
33 write.csv(expr, "GSE9150_normalized_expr.csv")
34 # 4. QC Plots
35 pdf("GSE9150_QC.pdf", width = 10, height = 6)
36 par(mfrow = c(1, 2))
37 boxplot(expr, main = "Boxplot of Normalized Expression", las = 2, col = "lightblue")
38 pca <- prcomp(t(expr), scale. = TRUE)
39 plot(pca$x[,1], pca$x[,2], col = pheno$Group,
40      pch = 19, xlab = "PC1", ylab = "PC2", main = "PCA Plot")
41 legend("topright", legend = levels(pheno$Group), col = 1:2, pch = 19)
42 dev.off()
43 # 5. Differential expression analysis
44 design <- model.matrix(~ Group, data = pheno)
45 fit <- lmFit(expr, design)
46 fit <- eBayes(fit)
47 deg <- topTable(
48   fit,
49   coef = "GroupDustMite",
50   number = Inf,
51   adjust.method = "BH",
52   sort.by = "P"
53 )
54 # Filter more significant DEGs
55 deg_filtered <- subset(deg, adj.P.val < 0.05 & abs(logFC) > 1)
56 # Save results
57 write.csv(deg, "GSE9150_DEG_FullList.csv")
58 write.csv(deg_filtered, "GSE9150_DEG_Significant.csv")
59 cat("\n✅ Analysis complete.\n")
60 cat("Full DEG list saved as: GSE9150_DEG_FullList.csv\n")
61 cat("Significant DEGs saved as: GSE9150_DEG_Significant.csv\n")
62 cat("QC plots saved as: GSE9150_QC.pdf\n")
63
```

CODE A.1 — Screenshot of R code for GSE9150

R pipeline for GSE9150 (dust allergy) — GEO download, RMA normalisation, PCA quality control, and limma differential expression analysis

```

1 # Install required packages if not installed
2 if (!requireNamespace("GEOquery", quietly = TRUE)) install.packages("GEOquery")
3 if (!requireNamespace("dplyr", quietly = TRUE)) install.packages("dplyr")
4 if (!requireNamespace("readr", quietly = TRUE)) install.packages("readr")
5 library(GEOquery)
6 library(dplyr)
7 library(readr)
8 # ==== 1. Load DEG file ====
9 deg_file <- "GSE9150_DEG_Significant.csv" # Change path if needed
10 deg <- read_csv(deg_file)
11 # Check first few rows
12 cat("First rows of DEG file:\n")
13 print(head(deg))
14 # ==== 2. Download GPL570 annotation ====
15 gpl <- getGEO("GPL570", AnnotGPL = TRUE)
16 gpl_table <- Table(gpl)
17 # Inspect columns
18 cat("Columns in GPL table:\n")
19 print(colnames(gpl_table))
20 # ==== 3. Keep only Probe ID & Gene Symbol ====
21 # Find gene symbol column automatically
22 gene_symbol_col <- grep("Gene.?Symbol", colnames(gpl_table), ignore.case = TRUE, value = TRUE)[1]
23 annot <- gpl_table %>%
24   select(ID, all_of(gene_symbol_col)) %>%
25   rename(ProbeID = ID, GeneSymbol = all_of(gene_symbol_col)) %>%
26   mutate(GeneSymbol = ifelse(GeneSymbol == "" | GeneSymbol == "----", NA, GeneSymbol))
27 # Remove rows with NA gene symbols
28 annot <- annot %>% filter(!is.na(GeneSymbol))
29 # ==== 4. Merge annotation with DEG data ====
30 # Ensure probe column in DEG matches "ProbeID"
31 colnames(deg)[1] <- "ProbeID" # assuming first col is probe ID
32 deg_annot <- merge(deg, annot, by = "ProbeID")
33 # ==== 5. Collapse multiple probes per gene ====
34 deg_gene <- deg_annot %>%
35   group_by(GeneSymbol) %>%
36   summarise(
37     logFC = mean(logFC, na.rm = TRUE),
38     AveExpr = mean(AveExpr, na.rm = TRUE),
39     t = mean(t, na.rm = TRUE),
40     P.Value = min(P.Value, na.rm = TRUE),
41     adj.P.Val = min(adj.P.Val, na.rm = TRUE),
42     B = mean(B, na.rm = TRUE)
43   ) %>%
44   ungroup()
45 # ==== 6. Create strict & discovery lists ====
46 # Strict: adj.P.Val < 0.05
47 deg_strict <- deg_gene %>% filter(adj.P.Val < 0.05)
48 # Discovery: adj.P.Val < 0.1 OR (P.Value < 0.05 & abs(logFC) > 0.5)
49 deg_discovery <- deg_gene %>%
50   filter(adj.P.Val < 0.1 | (P.Value < 0.05 & abs(logFC) > 0.5))
51 # ==== 7. Save outputs ====
52 write_csv(deg_gene, "GSE9150_DEG_GeneLevel.csv")
53 write_csv(deg_strict, "GSE9150_DEG_Strict.csv")
54 write_csv(deg_discovery, "GSE9150_DEG_Discovery.csv")
55 cat("☑ Mapping, collapsing, and filtering completed.\n")
56 cat("Files saved:\n")
57 cat(" - GSE9150_DEG_GeneLevel.csv (collapsed)\n")
58 cat(" - GSE9150_DEG_Strict.csv (adj.P.Val<0.05)\n")
59 cat(" - GSE9150_DEG_Discovery.csv (adj.P.Val<0.1 or p<0.05 & |logFC|>0.5)\n")
60
61

```

CODE A.2 — Screenshot of probe mapping + Venn overlap R code for GSE9150. R code for probe-to-gene symbol mapping and cross-disease gene overlap analysis — 15 shared discovery genes identified

```
[ ]
import pandas as pd
# =====
# LOAD FILES
# =====
rna = pd.read_excel("/content/RNA.SINUSITIS.CSV.xlsx")
prot = pd.read_excel("/content/PROTEIN.SINUSITIS.CSV.xlsx")
meta = pd.read_excel("/content/METABOLITE.SINUSITIS.CSV.xlsx")
# =====
# CLEAN COLUMN NAMES
# =====
for df in [rna, prot, meta]:
    df.columns = df.columns.str.strip()
# Rename consistently
rna = rna.rename(columns={'Pathway':'pathway', 'log10(q)':'log10_q'})
prot = prot.rename(columns={'Pathway':'pathway', 'log10(q)':'log10_q'})
meta = meta.rename(columns={'Pathway':'pathway', 'log10(q)':'log10_q'})
# =====
# CONVERT TO SCORE
# =====
rna['RNA'] = -rna['log10_q']
prot['Protein'] = -prot['log10_q']
meta['Metabolite'] = -meta['log10_q']
rna = rna[['pathway', 'RNA']]
prot = prot[['pathway', 'Protein']]
meta = meta[['pathway', 'Metabolite']]
# =====
# MERGE ALL OMICS
# =====
merged = rna.merge(prot, on='pathway', how='outer') \
    .merge(meta, on='pathway', how='outer') \
    .fillna(0)

# GROUP BY CATEGORY
grouped = merged.groupby('Category').agg({
    'RNA': 'sum',
    'Protein': 'sum',
    'Metabolite': 'sum',
    'Combined_Score': 'sum',
    'Consistency': 'max'
}).reset_index()
# =====
# OVERLAP %
# =====
total = len(grouped)
common = len(grouped[grouped['Consistency'] >= 2])
overlap_percent = (common / total) * 100
# TRUE overlap (same pathway in multiple omics)
merged['Pathway_Consistency'] = (
    (merged['RNA'] > 0).astype(int) +
    (merged['Protein'] > 0).astype(int) +
    (merged['Metabolite'] > 0).astype(int)
)
true_overlap = merged[merged['Pathway_Consistency'] >= 2]
```

CODE A.3 — Screenshot of Python cross-omics composite score code

Python code for quantitative cross-omics pathway integration and composite convergence score computation

```

▶ import pandas as pd
# =====
# LOAD PAN DATA
# =====
pan = pd.read_excel("/content/pan_transcr.xlsx")
pan.columns = pan.columns.str.strip()
pan = pan.rename(columns={'Pathway': 'pathway', 'Log10(q)': 'log10_q'})
# =====
# SCORE
# =====
pan['Pan_Score'] = -pan['log10_q']
# =====
# STRONG FUNCTIONAL CONVERGENCE
# =====
def categorize(p):
    p = str(p).lower()
    # Cytokine signaling
    if any(x in p for x in [
        'interleukin', 'cytokine', 'jak', 'stat', 'tyrobp',
        'immune signaling'
    ]):
        return 'Cytokine signaling'
    # Immune activation
    elif any(x in p for x in [
        't cell', 'lymphocyte', 'immune response', 'cell activation'
    ]):
        return 'Immune activation'
    # Inflammation
    elif 'inflamm' in p:
        return 'Inflammation'
    # Host-pathogen
    elif any(x in p for x in [
        'bacter', 'infection', 'pathogen'
    ]):
        return 'Host-pathogen interaction'

```

CODE A.4 — Screenshot of Python code for pan-grouped pathway analysis. First section of full script shown. Complete code available on request.

```

import pandas as pd
# =====
# LOAD DATA
# =====
pan = pd.read_csv("/content/pan_grouped.csv")
multi = pd.read_csv("/content/final_multiomics.csv")
# Clean column names
pan.columns = pan.columns.str.strip()
multi.columns = multi.columns.str.strip()
# =====
# RENAME FOR CONSISTENCY
# =====
pan = pan.rename(columns={'Pan_Score': 'Pan_Score'})
multi = multi.rename(columns={'Combined_Score': 'Multi_Score'})
# =====
# MERGE (BRIDGE STEP)
# =====
bridge = pd.merge(multi, pan, on='Category', how='outer').fillna(0)
# =====
# NORMALIZE (IMPORTANT FOR FAIR COMPARISON)
# =====
bridge['Pan_norm'] = bridge['Pan_Score'] / bridge['Pan_Score'].max()
bridge['Multi_norm'] = bridge['Multi_Score'] / bridge['Multi_Score'].max()
# =====
# DIFFERENCE & AGREEMENT
# =====
bridge['Difference'] = bridge['Pan_norm'] - bridge['Multi_norm']
bridge['Agreement'] = 1 - abs(bridge['Difference'])
# =====
# CLASSIFY RELATIONSHIP
# =====
def interpretation(row):
    if row['Pan_norm'] > 0.3 and row['Multi_norm'] > 0.3:
        return "Common core pathway"
    elif row['Pan_norm'] > 0.3 and row['Multi_norm'] <= 0.3:
        return "Transcript-dominant"

```

CODE A.5 — Screenshot of Python bridge step code

Python code for bridge step comparison between pan-transcriptomic and sinusitis multi-omics pathway themes. First section of full script shown. Complete code available on request.

```

# =====
# INSTALL
# =====
!pip install GEOparse
# =====
# IMPORTS
# =====
import GEOparse
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (accuracy_score, roc_auc_score,
                             classification_report, roc_curve)
from sklearn.feature_selection import SelectKBest, f_classif
# =====
# LOAD DATA
# =====
gse = GEOparse.get_GEO("GSE65204", destdir=".")
expr = gse.pivot_samples('VALUE')
gpl = list(gse.gpls.values())[0]
annot = gpl.table[['ID', 'GENE_SYMBOL']]
annot.columns = ['ID_REF', 'GeneSymbol']
annot = annot.dropna()
annot['GeneSymbol'] = annot['GeneSymbol'].str.split('///').str[0]
# =====
# MERGE
# =====
expr = expr.reset_index()
merged = expr.merge(annot, on='ID_REF')
merged = merged.groupby('GeneSymbol').mean(numeric_only=True)
# remove bad genes
merged = merged[~merged.index.str.startswith('LOC')]
merged = merged[merged.index.str.len() < 10]
df = merged.T

```

CODE A.6 — Screenshot of Python ML Part 1 code (genome-wide RF + LR). Python code for Part 1 genome-wide machine learning classification using Random Forest and Logistic Regression on GSE65204. First section of full script shown. Complete code available on request.


```

▶ # SINGLE clean plot
plt.figure(figsize=(6,5))
plt.barh(top_genes['Gene'], top_genes['Importance'], color=top_genes['Color'])
plt.gca().invert_yaxis()
plt.title("Top 15 Feature Importance (RF)")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()

# =====
# PART 2 (USE ORIGINAL X – FIXED)
# =====
biomarkers = [g for g in biomarkers if g in X.columns]
X_train_bio, X_test_bio, y_train, y_test = train_test_split(
    X[biomarkers], y, test_size=0.2, stratify=y, random_state=42
)
rf_prob2 = evaluate(rf, "PART2 - RF", X_train_bio, X_test_bio)
lr_prob2 = evaluate(lr, "PART2 - LR", X_train_bio, X_test_bio)
# =====
# ROC PART2
# =====
plt.figure(figsize=(7,5))
for prob, name in zip([rf_prob2, lr_prob2], ["RF", "LR"]):
    fpr, tpr, _ = roc_curve(y_test, prob)
    auc = roc_auc_score(y_test, prob)
    plt.plot(fpr, tpr, linewidth=3, label=f"{name} (AUC={auc:.2f})")
plt.plot([0,1],[0,1], 'k--', linewidth=2)
plt.title("ROC Curve ")
plt.xlabel("FPR")
plt.ylabel("TPR")
plt.legend()
plt.grid(alpha=0.3)
plt.show()

```

```

▶ # =====
# BIOMARKER IMPORTANCE (ALL 4 FIXED)
# =====
rf.fit(X_train_bio, y_train)
bio_imp = pd.DataFrame({
    'Gene': biomarkers,
    'Importance': rf.feature_importances_
})
plt.figure(figsize=(6,2.5))
plt.barh(bio_imp['Gene'], bio_imp['Importance'], color='steelblue')
plt.gca().invert_yaxis()
plt.title("Biomarker Importance (RF)")
plt.tight_layout()
plt.show()

```

CODE A.7 — Screenshot of Python ML Part 2 code (4 biomarkers only)

Python code for Part 2 four-biomarker panel classification using CCL5, CCL20, IL7R, and IDO1.